

Bicycle Parking Facilities on Campus

Praxis I Design Recommendation

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Lisa Li, Yasser Sleiman, Yuhan Zhu

Team R04

1. Introduction

Biking is a popular method of transportation across campus. It is a low-cost way to travel long distances, incorporate exercise, and reduce greenhouse gases. However, people are still forced to think twice before locking their bikes along the streets. According to the UTSG Campus Police Services annual report, despite the campus police having taken great action [1] by trying to proactively go after potential bike thieves, there were still 66 bike thefts in 2016 [2]. The original design brief aimed to decrease bike theft by redesigning the current bike locking system on campus; the current issue with the bike racks and bike rings is that cyclists tend to tie down only one or two parts of their bike (i.e. one wheel and the frame) to the rack, allowing for other parts to easily be stolen. When such a theft occurs, even if the Campus Police is contacted, unless the bike is registered or has a tracking system, the chances of retrieving the bike are incredibly small [1].

By reframing the original brief, and through a combination of diverging and converging our ideas, we ended up with the following objectives and requirements. These were used to decide on our final alternatives, and to pick our final chosen design. All the alternatives are services meant to be provided by campus, and aim to reduce bike theft to a minimum. When deciding on alternatives, our two main concerns were limiting the number of bike thefts of any kind, and being able to alert Campus Police whenever a theft was occurring. Our key metrics then became “the number of exposed bike parts” and “number of alarm triggers”. Through the use of a Pugh chart and a Rating Matrix, we ended up picking our original “retractable chain” bike locking system as the best alternative.

The following design report will present the requirements and objectives that were used, describe our different alternatives, as well as outline how we came to our conclusion.

2. Requirements

2.1 Original design brief requirements:

Among the metrics mentioned in the original brief there were: cost, durability, and usability. Cost was concerned with installation and maintenance, durability was concerned with how many years the product would last, and usability was concerned with how “easy” it was to use something.

For cost, we decided to remove it entirely as a requirement. The original brief’s scope was looking for a solution that could be either a product or service, but in our reframing, we decided to just focus on services. This is because the Campus Police officer we interviewed, the head of the bike theft department, mentioned how the cost of bike locks and the small budget students have resulted in people, more often than not, using locks that could easily be broken, instead of sturdier and more expensive ones [1]. This is solved by having a service as a solution, as it will not be expensive for the student, who will not have to pay for its installation. The University of Toronto has already invested resources in trying to

fight bike theft. An example of this is the “Bait Bike” program in 2012, where bikes with GPSs were purposely poorly locked and used as bait in order to find a catch thieves [3], so it should not be difficult for them to cover the cost of a new service.

Durability ended up being replaced by repairability, as it is more realistic that when the service/product ends up breaking it would not be due to its age or exposure to the elements, but due to a thief trying break it in order to steal a bike. Minimizing the ways a person can break the bicycle parking facility is more important than minimizing how long it takes before it breaks by itself.

As for Usability, in the original report it mentioned that the service/product had to be easy to use, as in anyone could use it without really needing to learn how. Although we addressed usability, we did not restrict ourselves to incredibly simplistic designs; as Steve Krug, a usability expert, puts it: using new conventions is acceptable if it “adds so much value that it’s worth a small learning curve” [4]. We believe that if people want to increase the security of their bike, learning how to use a new system is a small price to pay.

2.2 Revised requirements:

2.2.1 High level objectives (HL)

1. Reduce the amount of thefts of any kind by securing more parts of the bike
2. Make sure the Campus Police is alerted whenever a theft occurs

2.2.2 Detailed objectives (DO)

Find a new bicycle parking facility that:

1. Deters thieves from wanting to steal the bike by not giving any opportunity to do so
2. Can be repaired if damaged by a thief trying to break the system
3. Fits in the current space on campus, and is efficient in the way it stores the bikes (i.e. can fit as many bike as possible within a given space)
4. Fits most types of bikes, regardless their unique characteristics (ex: size, shape,...)
5. Is convenient to the user (should not take too much time or effort from the user)

2.2.3 Metrics

1. Number of exposed bike parts (HL1) (Key metric)
2. Number of alarm triggers (HL2, DO1) (Key metric)
3. Time taken to contact Campus Police (HL2)
4. Number of ways the system can be broken (DO2)
5. Site area taken up per bike lock (DO3)
6. Number of dimensional restrictions caused by the system (DO4), (a dimensional restriction being when a bike cannot fit into the system do to its irregular shape)
7. Ratio of number of parts secured to time needed to lock it, measured in seconds (DO5)

2.2.4 Constraints

1. The solution must be a system that is to be provided by campus rather than a product that can be bought by an individual
2. The system must be designed for short-term parking, as defined by the Toronto Guidelines for the Design and Management of Bicycle Parking Facilities [5]
3. The new system must fit into areas that are currently allocated for bicycle parking
4. Time needed to lock/unlock system cannot exceed 2 minutes (DO5) in order to keep it from being inconvenient

2.2.5 Criteria

1. Prefer less exposed bike parts
2. Prefer more alarm triggers
3. Prefer less time taken to contact Campus Police
4. Prefer less ways the system can be broken
5. Prefer less site area taken up per bike
6. Prefer less dimensional restrictions
7. Prefer higher ratio, meaning more parts are secured in less time

3. Design Alternative Descriptions

3.0 Original design brief alternatives:

Two reference designs were included in the design brief: the “Manifesto Bike Hanger” and “Trackr”:

- The Manifesto Bike Hanger is a ferris-wheel-like bike storage system, it can store up to 15 bicycles per hanger, and is electricity-free. However, due to it being electricity-free, the rotating mechanism requires the user to pedal on a stationary bike (shown on the bottom left in the image), and after watching a Youtube video [6], the system violated Constraint 4.
- Trackr is a small GPS that is connected to your phone. It is, however, a product, therefore violating Constraint 1.

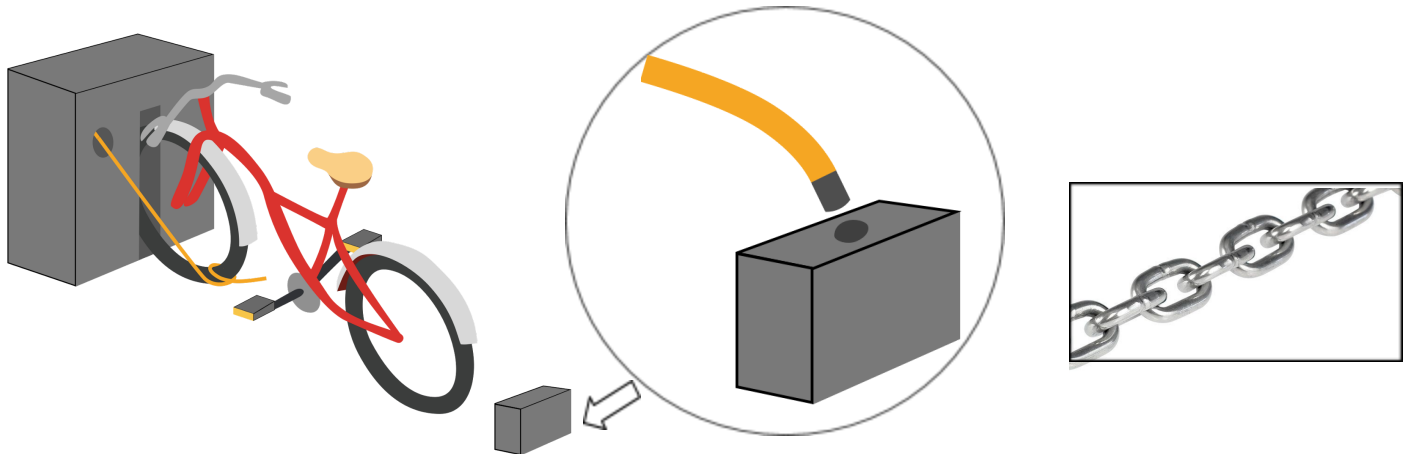


On the left:
Manifesto Bike Hanger [7]

On the right: Trackr [8]

3.1 Retractable chain bike station (Final recommendation)

This is an invention of our own; it uses a retractable chain (shown in orange for clarity) that can be wrapped around every part of the bike and then locked into place in a slot behind it. If applied correctly it can fit around different sized bikes and protect each and every part of the bike. We have implemented a security system that gets triggered whenever the chain is cut or when the lock is broken. When this happens a silent alarm is then sent to Campus Police and a loud siren goes off to alert bystanders of the thief's presence.



The retractable chain uses a mechanism similar to that of a seat belt, with a retractor that gets triggered by the movement of the chain and a pretensioner that helps hold the chain in place [9]. Just like with a seatbelt, a retractor mechanism is implemented, this means the chain can be drawn and then, when released, respool itself, this makes it so that when the chain is tied around the bike it will tighten around it, and if the chain is cut it will completely retract causing the system's alarms to trigger. A pretensioner is also implemented, this system can stop the chain's movement when someone attempts to pull on it while it is being used to lock a bike.

After some research on chains and the ways they can be broken or cut, we ended up picking a hardened steel chain with a diameter of 16 mm, which is big enough that it cannot easily be cut by any manual bolt croppers and hacksaws [10] This was also verified by the bike store owner of "Bike on Wheels" whom we spoke with [11].

Originally, we chose to use TCards to operate the lock, meaning that the lock was only available to students of the university, however we needed this system to be accessible by all, since in the guidelines 2.1., it is specified that as a short-term bicycle parking space, it should be available for public use [5]. After more converging and diverging, this led us to make the choice to also include Presto cards allowing for more people to have access to the system. Below was our thinking process: System 1 was used to come up with as many ideas as



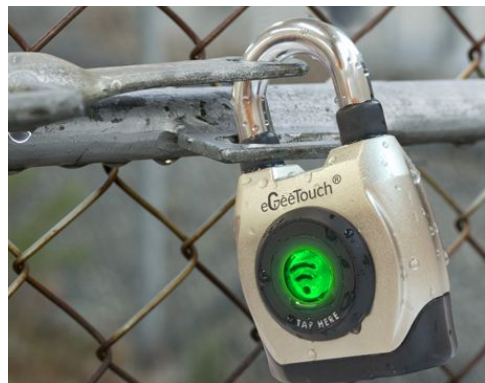
possible in 5 minutes, then System 2 was used to analyse the ideas, which eventually led to the decision of a Presto Card. Ideally, the system would be free to use and the Presto Card would only be used for verification purposes when locking and unlocking the system. However, this does the give the school the ability to charge non-students, or even students, a small fee to use the system [12].

Presto Card - Not everyone has one - People who live in the city do (and tend to be the ones riding bikes downtown) - Can be used to charge money (if the university decides to do so)	Credit Card - Not everyone has one - Privacy concerns - Seems very difficult to implement - Can be used to charge money (if the university decides to do so)	Text service (similar to how Tindr texts you a password everytime you want to log in), then you input pin into machine - Makes system too complex - Hacking becomes a new theft issue
Set new password for each use - Hacking becomes a new theft issue - May forget password	App (similar to Dropbike) - Not everyone has data - Requires building an app	

When we presented the design to a bike shop owner in downtown Toronto, he agreed that the system would work and prevent bike theft as it takes into consideration all of the factors that make a bike locking system effective (ex: no exposed bike parts) [11].

3.2 Velodome Shelters Wire Bike Shelters [13]

This “bike cage” provides an extra level of security as the bike is closed off. The user is required to open the gate, hang their bike vertically, and lock the bike to the stand. The bike and door are locked with the Weatherproof, Outdoor Smart Padlocks, which only needs Bluetooth connection from a phone to lock or unlock it. Bikes are vertically hung for more efficient use of the site area. Alarms are activated if either the door or the lock within the cage are broken.



Weatherproof, Outdoor Smart Padlocks by eGeeTouch.

3.3 Urban Racks Urban Double Stacker Rack

The Urban Double Stacker [14] is a two tier park parking system, where the bike is rolled into position and where the wheels can be locked into place. To place a bike on the top rack, a pneumatic lift assist enables anyone of any agility to use it. The Weatherproof, Outdoor Smart Padlocks are also used to lock the bike to the stand [15].



3.4 Bike rings

There are around 1600 bike rings across the St. George Campus [16]. Although the City of Toronto provides these rings upon request, the rings on campus are owned by the university. The design of the university's, identical to the City's, is a sleeved ring bolted onto a post. The incredibly simplistic design of the bike ring allows for bikes to be locked in a large number of ways, though there is the issue that some of those ways may be incorrect and may lead to the bike being easily stolen.



4.0 Design Evaluation

4.1 Rating matrix

<u>Objective</u>	Metric	Alternative 1: Retractable chain	Alternative 2: Wire Bike Shelter	Alternative 3: Urban Double Stacker Rack w/ digital locks	Alternative 4: Bike rings
<u>Prevention</u>	1. Number of exposed parts (Key metric) System 1	0 (if used properly along the entire bike)	0 (as long as cage remains closed)	5 {Handlebar, brake, seat, frame, pedal}	6 {Handlebar, brake, 1 wheel, seat, frame, pedal}
<u>Deterrence</u>	2. Number of alarm triggers (Key metric) System 1	2 {Chain broken and retracted, lock broken}	2 {Door is broken into, bike lock within is broken}	1 {Lock broken}	0
<u>Deterrence</u>	3. Time taken to contact Campus Police System 2	Instantly (due to a silent alarm sent directly to them)	Varies from when bike is stolen to when owner realizes and calls the Campus Police	Varies from when bike is stolen to when owner realizes and calls the Campus Police	Varies from when bike is stolen to when owner realizes and calls the Campus Police
<u>Repairability</u>	4. Number of ways the system can be broken System 1	2 {Chain, alarm system}	3 {Door, lock, cage}	2 {Lock, rack}	0
<u>Availability</u>	5. Site area per bike lock System 2	1.2 m² (0.6 m by 2 m)	1.3 m² (1 m by 1.3 m)	0.9 m² (0.45 m by 2 m)	0.04 m² (0.4 m by 0.1 m)
<u>Universality</u>	6. Dimensional restrictions of bike System 1	1 {Length of bike}	3 {Length, width, and height}	1 {Length}	0
<u>Ease of use</u>	7. Ratio of number of parts secured to time needed to lock it System 2	7 parts : 45 sec (minimum time: 7 sec if you chose to lock up only one part)	7 parts : 31 sec (minimum time: 31 sec because the bike can only be locked up as a whole)	2 wheels : 31 sec (minimum time: 15 sec if you chose to lock up one wheel)	1 wheel or frame : 5 sec (minimum time: 5 sec)

The parts of a bike that we considered could be stolen were: the handlebar, the brakes, both wheels, the seat, the frame, and the pedal, for a total of 7 parts. Accessories such as baskets were not included. The number of bike parts was used for M1 and M7.

The numbers for M1, M2, M4, and M6 were obtained from System 1.

The time for M3 was found with System 2 through research about silent alarms [17], and “instantly” was deemed a reasonable answer. As for Alternatives 2 and 3, the digital lock we found could not transmit an alarm to Campus Police, so it depended on when the bike was discovered stolen, identical to Alternative 4.

For M5 (site area), the dimensions of Alternative 2 and 3 were available on the products’ websites [13][14] and the dimensions of Alternative 4 were in Appendix A. As for the recommended solution, the best guess was taken. The length was taken to be the same as Alternative 3 as they both contain the entire length of a bike. The width was estimated to be wider than Alternative 4, as the main body of the system must contain a long length of chain as well as the mechanics for the TCard and Presto Card swipes.

For M7 (ratio), an actual bike was used to prototype the time needed. Simulations of all 4 alternatives were created and 3 trials were timed. The final times used in the ratio was the average of the 3, then rounded to the nearest second. For Alternative 1, a thick rope was used to loop around all 7 parts of the bike. For Alternative 2, a simple combination lock was used as a prototype of the digital lock, a small room with a door was used as a simulation for the cage, and the bike was propped up against the wall in place of hanging it from the hook. For Alternative 3, a snowboard was used in place of the rack to simulate pulling down the rack. A short set of stairs helped produce the incline and the bike was wheeled the top and locked. Timing Alternative 4 was merely closing the lock around a wheel. We believe that these prototypes give a fair estimation to the time it takes to lock a bike for each alternative.

	Alternative 1: Retractable chain	Alternative 2: Wire Bike Shelter	Alternative 3: Urban Double Stacker Rack w/ digital locks	Alternative 4: Bike rings
Trial 1	46.34	25.57	28.14	4.99
Trial 2	43.03	39.89	32.77	5.20
Trial 3	42.58	28.45	31.20	5.08
Average	43.98	31.30	30.70	5.09
Rounded	44 seconds	31 seconds	31 seconds	5 seconds

Table 1: Experimental data for M7; time needed to lock the bike using prototypes of each alternative.

Combination lock used as a prototype



Prototype of Alternative 1

Green rope is used in place of chain.



Prototype of Alternative 2

Bike was propped up against the wall. Opening and closing the door was also included in the time.



Prototype of Alternative 3

Bottom of a snowboard was used as the rack and the bike was rolled up to the top of the stairs.



4.2 Pugh Chart

Metric	Alternative 1: retractable chain	Alternative 2: Wire Bike Shelter	Alternative 3: Urban Double Stacker Rack w/ digital locks (Reference)	Alternative 4: Bike rings
1. Number of exposed parts (Key metric)	+	+	same	-
2. Number of alarm triggers (Key metric)	+	+	same	-
3. Time taken to contact Campus Police	+	same	same	same
4. Number of ways the system can be broken	same	-	same	+
5. Area taken up per bike lock	-	-	same	+
6. Dimensional restrictions of bike	same	-	same	+
7. Ratio of number of parts secured to time needed to lock it	+	+	same	+

The results of the Pugh charts were based off the data from the Rating Matrix. Alternative 3 was chosen as the reference because it was a good middle ground where the alternatives have a balance of betters and worses. From this, we see that although Alternative 4, the current system, scores better than the reference in M4, 5, 6, and 7, but scored worse in the 2 key metrics. The key metrics were chosen as the key metrics because they were the main issues with the current system.

Alternative 1 and 2 both scored better in the key metrics compared to Alternative 3. For M3, Alternative 3 had the quickest alert time as the other 3 all varied, depending on when the owner returned

to the bike as there was no way to alert Campus Police once the system was broken into. When comparing the alternatives with M4 and M5, Alternative 4 had the least ways to be broken as well as the smallest area. However, this is due to its simple design and may be the reason why bike parts are so easily removed. For M6, Alternative 4 allowed for more types of bikes to be parked compared to the other alternatives. However, this metric is not as important as the other ones as all the alternatives will fit standard bikes, and M6 was included with bikes of extreme physical properties in mind. Lastly for M7, all 3 alternatives had a better ratio of number of parts secured to time needed to lock it than the reference, which may indicate that Alternative 3 was the least user-friendly system.

Based on the Pugh Chart, Alternative 1 seemed to be the highest ranking alternative.

4.3 Pairwise Comparison

	Retractable chain (A)	Wire Bike Shelter (B)	Urban Double Stacker Rack with digital locks w/ digital locks (C)	Bike rings (D)
Retractable chain (A)	-	A	A	A
Wire Bike Shelter (B)	-	-	B	B
Urban Double Stacker Rack with digital locks w/ digital locks (C)	-	-	-	D
Bike rings (D)	-	-	-	-

The data from the rating matrix was used to compare each pair of alternatives. When comparing A and B, many results from the rating matrix were identical, including the key metrics. However, A had more alarms than B and is the only alternative that is able to immediately contact Campus Police, which fulfills more important objectives than the metrics where B scored better than in A. When comparing A with C and D, A scored better in both key metrics compared to C and D, as well as other objectives, leading A to rank as the best design in this pairwise comparison.

4.4 Evaluation of Results

As seen in the previous Pugh and Pairwise comparison chart, the retractable chain bike lock station is the overall best among the four alternatives. By comparing it to the others, it becomes apparent that its characteristics make it the most suitable option within our objectives and constraints. This design protects the entirety of the bicycle and can alert the Campus Police the most effectively, not only that, it is space efficient, is relatively easy to use, is safe. These are all reasons why we decided to pick the retractable chain bike lock station as the alternative we are going to move forward with.

5. Next Steps:

Having decided on which design we are going to be focusing on for the rest of the project, what remains now is to take a much closer look at the different parts and mechanisms involved in making a functioning realistic model of the retractable chain bike lock station. Among the aspects of the design that should be studied, we have: the alarm system; the chain; the chain retraction system; the card identification system.

For the alarm, some testing should be done to find the exact conditions for the alarm to set off, i.e. how much should the chain retract for the alarm to go off, ... Not only that but we should look into the details of how loud the siren should be in order to alert bystanders, but not disturb the surrounding buildings too much. The best method that could be used to send the silent alarm to Campus Police should also be chosen (online; radio; ...).

For the chain, we should find if the dimensions or weight of the chain might cause problems for some users. The coating material of the chain, that stops the bikes from being scratched or damaged when it tightens and that protects the chains from the elements, should also be chosen (as recommended in *2.3.1. Rack Design* of [12]).

The retraction system will require the most study if an existing model that fits our requirements cannot be found. We would need to design a system that could both reel in an entire chain rapidly, while also having no chance of damaging bikes or causing some form of injury. It should also integrate a some form of sensor for the alarm system to trigger.

Lastly for the card identification system we should work in collaboration with the university, more specifically the people who installed the TCard scanners all over campus, to learn about the technology behind it, such that we may be able to apply it ourselves. We should also contact a representative of Presto to see if there are any restrictions or any permissions we should have or know about implementing their system into our scanner.

Analysing Bicycle Parking Systems at University of Toronto's St George Campus

November 4, 2017

Introduction

Cycling has become a common mode of transportation of the University of Toronto's Saint George Campus (UTSG) used by campus staff, service providers, and students alike. Beneficial to one's health [1], faster than walking [2] [3] and more environmentally friendly than automobiles [4], cycling offers a variety of benefits that users enjoy. With every bicycle on campus, a place to keep it when not in use is required, thus causing the need for biking parking systems. This need has an impact on many of UTSG's service providers including campus police, maintenance staff, and UTSG cyclists.

One of the major issues facing cyclists is the danger of bike theft. In 2016, University of Toronto Campus Police, a major service provider, reported 66 bike thefts [5] throughout the year. In 2017, 18 thefts have been reported in the month of October alone [6]. The primary method of theft prevention is through the system of bike posts and racks throughout campus, onto which cyclists can lock their bikes. Statistics show that while bicycle theft rates are decreasing, [Appendix D] current bicycle parking systems at UTSG are poorly designed for security despite how common cycling is on campus.

This report aims to establish requirements by which engineered solutions to bicycle theft at UTSG can be compared against. It analyzes and evaluates the current bicycle parking system at UTSG in terms of their impact on campus police, UTSG cyclists, and UTSG maintenance staff. It then demonstrates that existing bicycle security solutions such as bike hangers and Trackr are not appropriate to UTSG. Flaws with the above are abstracted to develop a system of requirements, metrics, constraints, and criteria to frame the opportunity from security, usability, and maintainability design perspectives.

Bicycle Thefts at the University of Toronto: St. George Campus

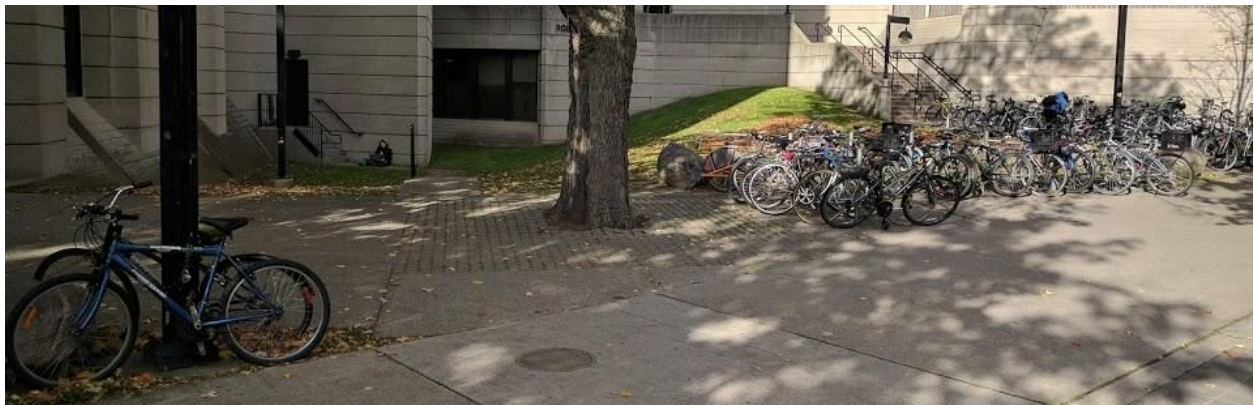


Figure 1: Many bikes were locked outside Robarts Library, 2017/10/30 10:30:00. Notice the bikes locked to the street light.

Bike thefts have occurred all over St. George Campus. Locations for bike theft include the bike rings outside Robarts and the rings outside the Earth Sciences building. The reason thefts are common at these locations might be because of the sheer quantity of bikes parked there (see Figure 1). Additionally, the Earth Sciences building might be a common target due to the lesser volume of people and cars that pass by that area, compared to Bahen for example. Bike thieves are less likely to strike in highly populated areas [C], but it is not impossible [D]. If a thief steals a whole bike, there is nothing to observe, so it is only possible to observe cases where bike parts are stolen. Most observable cases of theft involve stolen wheels or seats, as shown in Figure 2 [D].



Figure 2: Bicycle U-locked to a bike ring is missing a wheel.

The most common current solution to bike theft is bike locks and bike rings. Standard bike locks and rings are simple, ubiquitous, and universal. However, there are issues with this solution in terms of security, usability, and maintainability. While U-locks and cable locks are convenient, they can only secure bicycles in the presence of bike racks and bike posts, all of which needing maintenance against weathering due to being outdoors [B,C,D]. Stakeholders on campus have had bikes stolen despite locking their bike onto bike rings [9], and bike rings provide minimal deterrence and protection against theft of entire bicycles and parts, which provides little information when campus police tries to track down and identify stolen bicycles [D]. Bicycles can be registered with the City of Toronto, but even this does little to prevent theft or assist recovery. The outdoor posts show signs of weathering which compromises its appearance as a secure bicycle parking system [B].

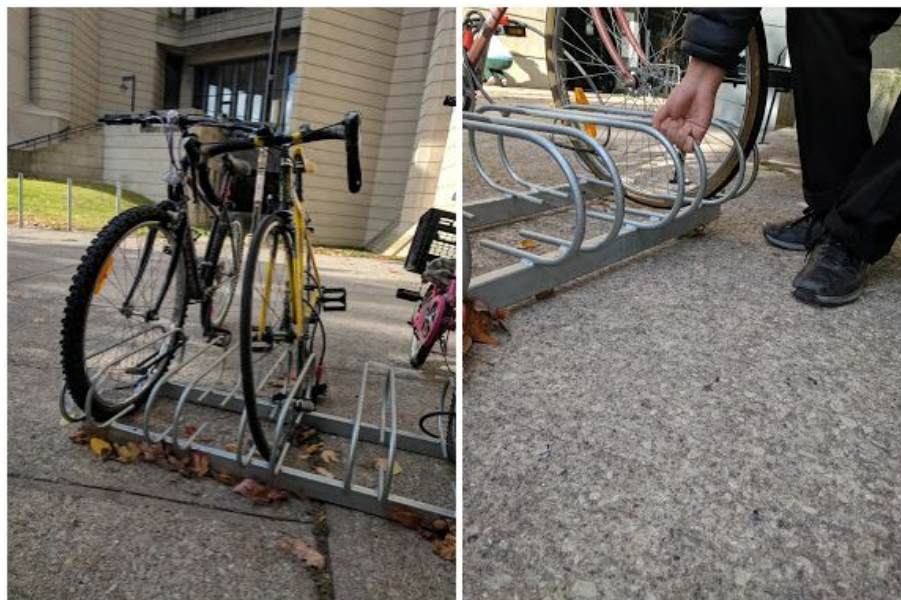


Figure 3: Wheel rack in front of Robarts Library with locked bicycles. U-lock on leftmost bike secures only the wheel, and the rack can be lifted from the ground.

Wheel racks are also used on campus, however these are easily removed from the ground thus reducing their efficacy as a security solution and increasing the likelihood of theft [D]. They also make it difficult to connect the frame of your bike to the stand, thus making your wheels safe but not the frame of your bike.

Bike Share bicycle sharing services are also located on campus. These systems have a locking stand for rental bikes, however they do not allow for much freedom of the user as users can only access the bikes in certain places, namely where the stands are located. For example, these may be present all around campus but it would be difficult to build a system in which these stands would be easy to access everywhere. The stands securely lock Bike Share bicycles, but these cannot be used to secure personal bikes.

Current bicycle parking implementations are therefore moderately usable, but fail to be secure. Signs of wear indicate that maintainability of the bicycle racks may also be an issue.

Stakeholders

a. Campus Police

Campus police has many patrols that watch over and protect the University. As such, they are an important service provider on campus. Their many duties include preventing, stopping, and reacting to bike theft. However, no matter how vigilant officers are, it is nearly impossible to keep watch over all bikes at all times. After a bike is stolen, the owner will often contact Campus Police. It is difficult to track down bike thieves, especially if only certain parts are stolen, such as wheels. Campus police adds the theft

to an activity report and summarizes all campus crimes in an annual report at the end of the year. The proposed solution should assist campus police by making it easier for them to prevent, stop, and react to bike thefts. Campus police also conduct anti-theft programs on campus such as Bait Bike in 2012, in which they placed GPS-tracked bicycles with minimal security across campus to be stolen and traced back to thieves. This led to a significant decrease in bicycle thefts the following year [D].

b. UTSG Cyclists

There has been an increase in the number of people using bicycles to replace other modes of transportation. These cyclists, some being service providers themselves like professors, are the users of the bicycle parking system. This means the usability of the system [C] and the safety directly affect the UTSG cyclists. They are concerned with the protection of their bicycles against theft and the convenience of the solution.

First years student Rui Gao in particular had his bike stolen during the day while it was chain locked [7].

c. Maintenance Staff at UTSG

The maintenance team at the university are responsible for caring for and repairing the issues around campus. These include managing the care and replacement of the bicycle parking stations provided at the different buildings at UTSG. The durability and maintainability of the bike racks used has a major effect on this service provider as this will determine the amount of repairs needed and the attention required.

Bicycle Parking Security Reference Designs

The following reference designs solve bicycle security issues in other locations, but are insufficient in addressing the needs of stakeholders at UTSG.

a) MANIFESTO Bike Hanger

Manifesto Design Labs plans to improve bicycle parking systems in Seoul with a vertical rotating hanger powered by a stationary bike [8]. It addresses a need for limited bicycle parking space with a slow rotating system that both makes bicycles easily identifiable and delays theft attempts.

However, there are many rotating gears and chains as part of the design that are more likely to fail and must be maintained more than current implementations. Moving systems by nature require lubrication which must be replenished regularly.

b) Trackr

Trackr is a personal asset tracking device that can be attached to belongings and traced via the Trackr application. Not only does it allow the bicycle and related thieves to be located, it deters thieves from

stealing both trackable and untrackable bicycles for fear of being traced. It effectively converts all installed to Bait Bikes used by consumers.

However, GPS tracking is a privacy concern. Trackr's privacy policy asserts that user information is analysed with Google Analytics [9], which sells information to third-parties for marketing purposes, creating a loophole that hides usage of customer data, bordering on violating the Canadian Privacy Act [10].

Requirements

- **Security:** The bicycle parking system should keep all attached property safe from attempted theft and damage [D].
- **Usability:** The system should be easy to use and serve its purpose to hold and secure bikes [C].
- **Maintainability:** The system should stand up to the everyday use as well as weathering due to the elements [B].
- **Cost:** The bike parking system should be affordable, or not very expensive to purchase and install.

Metrics and Criteria

1. Cost
 - a) Metric: Canadian dollar
 - b) Installation cost - In general, lower cost is preferred
 - c) Maintenance cost - In general, lower maintenance cost is preferred
2. Durability
 - d) Metric: Years - In general, longer lasting alternatives are preferred
3. Metrics and criteria of qualitative objectives are as follows:

Objective	Less desirable		More desirable
Ease of use [C]	Multiple parts; more complicated to use than bike locks and racks	About the same difficulty of use as current alternative	Extremely intuitive; easier to use than bike locks and racks
Availability of use [C]	Can only be used in limited locations; allows for less parking options than bike locks and racks	Can only be used if a bike rack or some other structure is available; similar availability to current solution	Can be used without the presence of existing structures

Universality [C]	Only usable for one height, length, etc. of bike	Can be used with most types of bikes to a certain degree; may be awkward to use with certain types of bikes; similar to current alternative	Can be used with nearly all types of bikes: mountain, dirt, etc.
Deterrence [D]	Wouldn't deter thieves at all; about the same effectiveness as cable locks, which can be cut with wire cutters	Would make thieves hesitate to act; possibly alerts bystanders to theft	Thieves will avoid attempting to steal the bike if this alternative is in effect; not only alerts bystanders but possibly leads authority to thief
Prevention [D]	About the same effectiveness as current solution; prevents theft of frame and possibly wheels; delays robbery, but unlikely to prevent intentional theft.	Protects both wheels and frame from being stolen; if breakable, takes long enough to break so that theft could be noticed	Prevents theft of frame and both wheels, as well as seat and/or helmet; requires excessive effort and time to break; thief would probably give up
Recovery [D]	Stolen bicycle leaves no evidence leading to criminal or recovery of bike; similar to current alternative	Provides some opportunity for the police to trace the theft and identify the owner of stolen property; moderate uncertainty of recovery; might have an unreliable trigger	Accurately tracks the stolen bike; easy to track down thief; reliable trigger, low chance of false alarm; possibly marks but doesn't harm the thief (in case of accidental trigger)
Maintainability [B]	Condition must be frequently checked and maintained to preserve efficacy.	Initially resists harsh conditions, but eventually weathers	Requires little to no maintenance to continue functioning under harsh outdoor conditions
Repairability [B]	Very difficult to	Vulnerable parts are	Parts can be easily

	acquire replacement parts; consists of many complicated parts; repair requires professional attention or thorough self-study and repair on the part of the user	easily removed and replaceable without disrupting rest of system; either composed a few complicated parts or many simple parts; can be fixed by user but requires tools that may not be readily available	acquired; made up of a few simple parts; can be fixed by user with common tools
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Constraints

1. The proposed alternative cannot violate the Canadian Privacy Act [10]
4. The proposed alternative must comply with the Canada Consumer Product Safety Act
5. If the proposed alternative is a personal device, it should not weigh more than the heaviest current options (i.e. chain locks $\approx$ 20lbs) [C]

Conclusion

Bike theft is a prominent issue at UTSG, and affects stakeholders and service providers including campus police, UTSG cyclists, and maintenance staff at UTSG. There are existing solutions, but all of them fall short when framed in at least one design aspect. Proposed alternative solutions must meet the constraints and satisfy the criteria illustrated above. Preventing bike theft is an opportunity to exercise engineering design by meeting various design objectives, including security, usability, maintainability/durability, cost, and privacy, among others.

Appendix A: Source Extracts

[1] H. Publishing, "The top 5 benefits of cycling - Harvard Health", Harvard Health, 2017. [Online]. Available: <https://www.health.harvard.edu/staying-healthy/the-top-5-benefits-of-cycling>. [Accessed: 04-Nov- 2017].

[2] "Bicycle statistics", Web.archive.org, 2017. [Online]. Available: <https://web.archive.org/web/20131212093813/http://subsite.kk.dk/sitecore/content/Subsites/CityOfCopenhagen/SubsiteFrontpage/LivingInCopenhagen/CityAndTraffic/CityOfCyclists/CycleStatistics.aspx>. [Accessed: 04- Nov- 2017].

“average travelling speed... is 15.5 km/h for cyclists”

[3] "psu_ped_summary.pdf", Westernite.org, 2017. [Online]. Available: https://www.westernite.org/datacollectionfund/2005/psu_ped_summary.pdf. [Accessed: 04- Nov- 2017].

“The average walking speed for younger pedestrians was 4.85 ft/second [5.32 km/h] and 4.33 [4.75 km/h] ft/second for older pedestrians.”

[4]"Environmental Benefits | Bike to Work Day", Youcanbikethere.com, 2017. [Online]. Available: <https://youcanbikethere.com/environmental-benefits/>. [Accessed: 04- Nov- 2017].

1. “Bicycling uses no fuel.
2. Bicycles take a lot less energy to make than a car.
3. Bikes don’t require toxic batteries or motor oil.”

[5] UTSG Campus Police Services, “Annual report,” 2016. [Online]. Available: <http://campuspolice.utoronto.ca/wp-content/uploads/2017/09/2016-Campus-PoliceAnnual-Report-University-of-Toronto-Affairs-Board.pdf>

REPORTABLE INCIDENTS

2016 Statistical Overview - St. George Campus

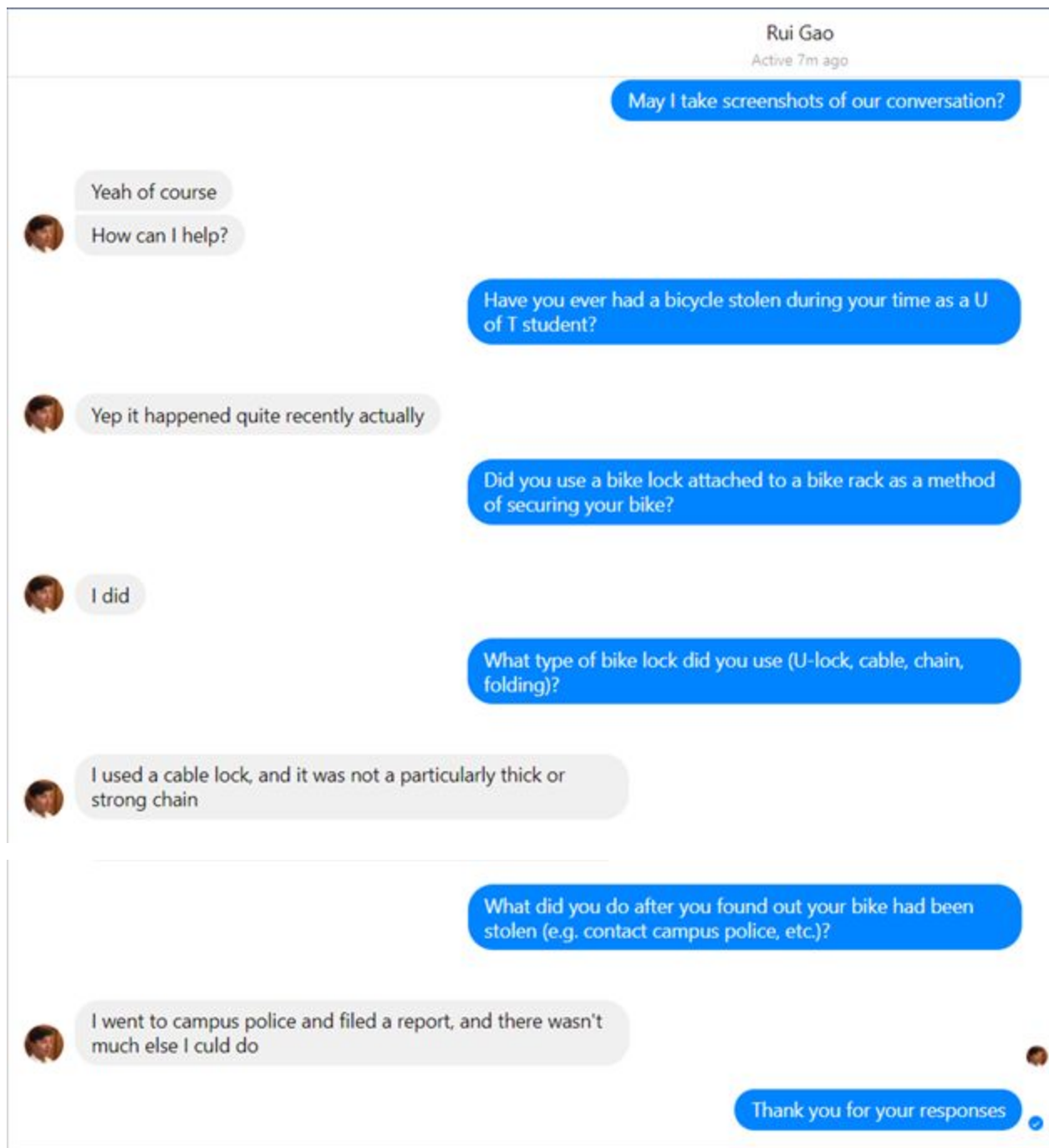
PROPERTY CRIMES	2016	2015	16vs15
Arson	1	1	0
Attempt Theft	3	3	0
Break and Enter	9	12	-3
Unlawfully in a Dwelling	1	0	1
Fraud/ False Pretences	6	3	3
Mischief over \$5000 Univ.	0	2	-2
Mischief under \$5000	112	140	-28
Mischief - Interfere lawful enjoyment of property	1	0	1
Theft over \$5000.00	4	5	-1
Theft under \$5000.00	228	297	-69
Theft under \$5000.00 Bicycles	66	75	-9
Possession Stolen Property	0	0	0
Classification Total	431	538	-107

[6] "Activity Reports for October 2017", Campuspolice.utoronto.ca, 2017. [Online]. Available: <http://campuspolice.utoronto.ca/events/2017-10/?tribe-bar-date=2017-10&tribe-bar-search=bicycle>. [Accessed: 04- Nov- 2017].

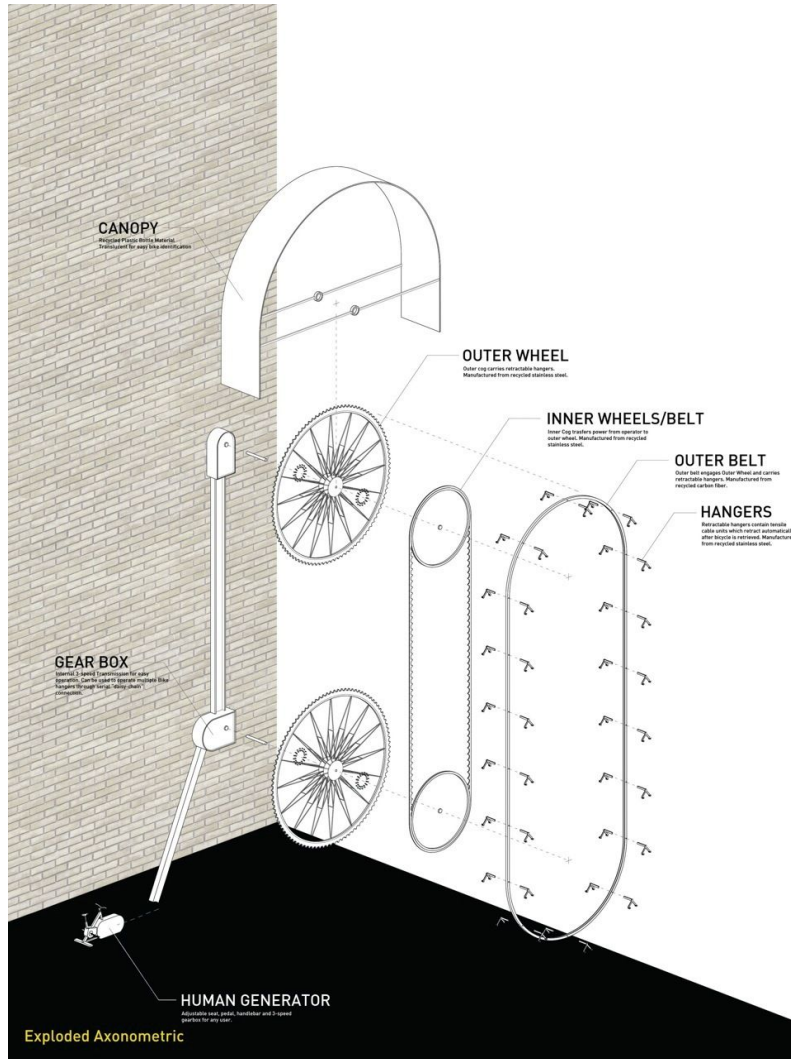
Activity Reports for October 2017

MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
25	Theft October 3 A person reported a stolen bicycle. UT17048819	27	28	29	30	1
2	Theft	4 Theft Theft	5	6	7	8
9	Theft	11 Theft	12	13	14	15 Theft
16	17 Theft	18 Theft	19 Theft Theft	20	21 Theft	22
23 Theft Theft	24	25 Theft	26 Theft Theft	27	28	29
30	31	1	2	3	4	5

[7] "Interview with Rui Gao," 2017.



[8] "MANIFESTO ARCHITECTURE", *Mfarch.com*, 2017. [Online]. Available: http://www.mfarch.com/designlab_detail_main.php?page_type=DESIGNLAB&detail_id=%3D%3DAUVZ0VWxGZXJFbaNVTWJVU. [Accessed: 04- Nov- 2017].



[9] "Privacy Policy - TrackR", TrackR, 2017. [Online]. Available: <https://www.thetrackr.com/privacy-policy/>. [Accessed: 04- Nov- 2017].

The Personal Data used for each purpose is outlined in the specific sections of this document.

Analytics

The services contained in this section enable the Owner to monitor and analyze web traffic and can be used to keep track of User behavior.

Google Analytics (Google)

Google Analytics is a web analysis service provided by Google Inc. ("Google"). Google utilizes the Data collected to track and examine the use of this Application, to prepare reports on its activities and share them with other Google services. Google may use the Data collected to contextualize and personalize the ads of its own advertising network. Personal Data collected: Cookie and Usage Data.

[10] O. Canada, "PIPEDA fair information principles - Office of the Privacy Commissioner of Canada", Priv.gc.ca, 2017. [Online]. Available: https://www.priv.gc.ca/en/privacy-topics/privacy-laws-in-canada/the-personal-information-protection-and-electronic-documents-act-pipeda/p_principle/. [Accessed: 04- Nov- 2017].

Principle 5 – Limiting Use, Disclosure, and Retention

Personal information shall not be used or disclosed for purposes other than those for which it was collected, except with the consent of the individual or as required by law. Personal information shall be retained only as long as necessary for the fulfillment of those purposes.

Appendix B: UTSG Bike Parking Site Inspection from the Perspective of Maintainability and Durability

Durability is the ability to exist for an extended period of time without significant deterioration in quality or value, while maintainability is the ability to keep in an existing state or to preserve from failure and decline[1,2]. Improving these factors in the design of the bicycle parking systems around the campus will improve the experience of the maintenance team at the University of Toronto as it will decrease the number of repairs needed on the system and improve the ease of needed maintenance.

At Robarts, the durability of the bicycle parking system must be improved as the floor racks are very easily damaged. As seen in figure [3], this rack has been disconnected from its anchoring system, meaning it can be lifted. By a set of standards found from [4], the rack must be able to resist force and not be displaced, which these racks clearly can be. The locking system to the ground therefore must be improved, as if the racks are constantly being moved and damaged this will create more issues for maintenance to repair. In the current system, the bike parking system does not have enough space to hold the number of bicycles being used on the campus, as seen in front of Robarts Library[5]. Increasing the capacity of the systems will reduce the use of other objects as anchors to lock to such as the light poles, which will decrease the damages to outside property. As well, since the bicycle rack is on the ground, this means in the winter it may get covered in snow, and we can see now in the fall they are getting covered in the leaves. The exposure to water will cause the metal to rust quicker, meaning they have to be replaced more often. Therefore, the placement of the racks reduces the durability as they are more susceptible to rust and more likely to deteriorate faster. The bars on the stands at Robarts are also not extremely thick, meaning they may be targets for attempts at theft. These may be bent or sawed through thus releasing the locks of the bikes and allowing them to be taken. This also destroys the rack, again creating another issue for the service providers at UofT.

The ring ranks around most of campus also must be improved with durability and maintenance in mind. As can be seen in figure [6], the stand does not withstanding the elements. It appears as if there is water-damaged or rust on the post. The stand also has many scratches meaning it probably often damaged by impact with bikes and locks. The rust will reduce the integrity of the stand, meaning it needs to be repaired more often. Rust is a form of corrosion that takes place when iron containing metals are exposed to water. It makes the metal weaker as it turns the strong iron or steel with flaky powder [7]. As well, the rust reduces the bond strength with the concrete thus making the stand easier to tip over and shift. Therefore, the rack is currently not very durable. The materials of the stand therefore may need to be changed as well as the water-resistance and drainage in the rack. This will help reduce the effects of the water on the stand thus improving its maintainability. As well, these racks are anchored by the cement below it. In the figure [8], one can see how the cement beneath the rack looks a different colour as it was newly changed to place the stand into the ground. This is a difficult process to do every time the stand needs to be replaced as it requires more tools, resources, and placing new stones or cement. Also, drilling into places where new stands are needed may also cause cracking in cement. This only creates more work for the maintenance team, and also increases the difficulty of the repairs. In many of the racks around campus paint is also chipping, which also creates more work for the maintenance team. In figure [4], one

can see that based on these guidelines the bicycle parking system must be weather proof, however from the rust and the chipping paint one can see that the stands are clearly not, and are thus not very durable.

Another issue with the current design that reduces the durability and maintainability of the current racks is the rust on the bolts of the parking stand [5]. This creates similar issues as above. The rust would make the bolts begin to weaken, thus making the stand easier to break. Finally, the lack of removable parts also makes the rack more difficult to maintain. The racks in front of Robarts do not have removable parts, as all the bars holding the wheels are welded to the main frame. This means if the bars are damaged the whole rack must be replaced as well as the bolts of the old rack must be removed from the ground as the whole thing is one piece. Similarly, the ring racks do not have many parts that can be separated and removed meaning if one has an issue, the whole rack must pretty much be replaced.

The issues affect the integrity of the bike stand and thus they affect the ability of the rack to fulfill its purpose and protect against theft. When the rack can no longer serve its purpose, they must be replaced or repaired, and if this occurs very frequently this will affect the maintenance team at the University. Therefore, one can see how the bike parking system on campus could benefit from a re-design with maintainability and durability in mind.

Appendix B Source Extracts

[1] "Definition of DURABLE", *Merriam-webster.com*, 2017. [Online]. Available: <https://www.merriam-webster.com/dictionary/durable>. [Accessed: 03- Nov- 2017].

[2] "Definition of MAINTAINABILITY", *Merriam-webster.com*, 2017. [Online]. Available: <https://www.merriam-webster.com/dictionary/maintainability>. [Accessed: 04- Nov- 2017].

[3]



[4] "Standards for Bicycle Parking System", *Fietsparkeur.nl*, 2017. [Online]. Available: <http://www.fietsparkeur.nl/normtekstengels.pdf>. [Accessed: 03- Nov- 2017].

7.3.1.1 Impact

The BPS should be vandalism-proof.

7.3.4.2 Synthetic parts

Synthetic materials used should be weather-resistant

[5]



[6]



[7] "Q & A: Effects of rust | Department of Physics | University of Illinois at Urbana-Champaign", *Van.physics.illinois.edu*, 2017. [Online]. Available: <https://van.physics.illinois.edu/qa/listing.php?id=494>. [Accessed: 03- Nov- 2017].

Rusting has a number of effects on metal objects. It makes them look orange and rough. It makes them weaker, by replacing the strong iron or steel with flaky powder. Some oxides on some

[8]



Appendix C: UTSG Bike Parking Site Inspection from the Perspective of Usability

This site inspection will analyze the usability of bike rings, locks, and rental bike stands within the University of Toronto St. George Campus (UTSG). Usability is defined as “the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use. [1].” The sites inspected consisted mostly of bikes locked to bicycle rings with various types of bike locks [2.1]. The standard U of T bike ring consist of a vertical metal post around 36 in in height, and a round metal ring around 16 in in diameter [2.10]. In this inspection, bike locks will be considered part of the site. In general, bike locks and rings are easy to use and readily available. Other sites included rental bike stands owned by Bike Share Toronto. These stands consist of rental bikes locked to a long bike rack, with bikes available for renting at a rate of \$7 per day. The main stakeholders at these sites are U of T students, professors, and other people who use bikes as a mode of transportation. Campus police are stakeholders and service providers that are responsible for dealing with bike theft at these sites.

There are 80+ locations of bike racks on and around UTSG [3.1]. Most of these consist of the aforementioned post and ring bike rack design, which are firmly mounted in the ground. Many of these bike rings were close to building entrances [2.2] [2.3], as recommended by the Bicycle Parking Guide for Business Owners & Cyclists in the City of Toronto [4]. This minimizes the amount of time it takes for cyclists to get to their destination after parking their bike. The height of the ring is around the same height as the top of most bikes’ wheels [2.4]. This allows cyclists to lock the frame and at least one wheel of their bike to the ring with different types of locks [2.5] [2.6] [2.7], which provides flexibility in terms of lock type and encourages cyclists to use the bike rings [4]. Usually two bicycles can be locked at a bike ring, on either side of the ring [2.1]. The ring provides sufficient stability for both bikes, with neither bike hindering the stability of the other, as per 7.1.2.3 of the FietsParKeur Bicycle Parking Systems Standard [5]. This makes it easier to park more bikes in an area without bikes toppling over. Vulnerable items on bikes will not come into contact with the bike ring, as per 7.1.2.2 of [5], which means these items are at lesser risk of breaking. Some rings were under areas that would provide shelter from rain and snow [2.3] as per [4], but many were located out in the open with no protection from the elements [2.1] [2.8]. If it rains or snows while a bike is parked, the cyclist will likely come back to a soaked or snow-covered vehicle. Most bike rings are visible from the road [2.10] [2.8], which makes it easier for cyclists to find a place to park [4]. Also, roads are generally high-use areas, and placing a bike ring so that it is visible from a road may deter potential thieves [4]. However, unless the user locks the wheels or seat of their bike, the bike ring does little to deter theft of bicycle parts. Additionally, if no rings are available, users will have to lock their bikes to other objects, such as trees, light posts, or handrails [2.2].

Different types of bike locks include U-locks, chain locks, folding locks, and cable locks. U-locks consist of a solid, curved metal section and a flat bar [6.1]. Since U-locks are rigid, they are more difficult to transport compared to chain and cable locks [6]. Also, because of their fixed shape, it is more difficult to find objects that you can secure your bike to [6]. Most U-locks have an inner width ranging from 2.64 in to 6.7 in [7.1] [7.2]. This would make it difficult to lock your bike around columns, thick trees, etc. However, this is less of a problem due to the abundance of bike rings around UTSG. U-locks are generally lighter than chain locks, with an average weight of around 2.9 lbs [7], compared to around 6.0

lbs for chain locks. The heaviest chain locks weigh upwards of 20 lbs [9]. Chain locks typically consist of a chain with a sleeve, and a large lock connecting both ends [6.2]. They are more flexible than U-locks and have lengths ranging from 21.5 in to 72.0 in [8.1] [8.2]. Folding locks are rigid like U-locks, but have several joints around which the lock can be folded [6.3] [6.4]. Because of this, they are easy to transport while being a practical length [6], and often come with carrying holsters that can be attached to bikes [6.4]. This means they will not hinder the cyclist during a ride [6]. Folding locks are also generally lighter than U-locks of a comparable level of security [6]. However, they are more difficult to use since they must be unfolded, and as a result it may take longer to secure one's bike [6]. Cable locks, like chain locks, are long and flexible, and additionally have a lighter weight [6.5]. However, cable locks are among the easiest bike locks to cut [6]. Some cable locks can be cut easily with standard wire cutters [9], so their usability benefits are outweighed by their drawbacks in security.

The other sites inspected were rental bike stands owned by Bike Share. There are 10+ Bike Share stands on and around UTSG [3.2]. To properly lock a rental bike, one must push the bike into the stand so that the probe on the front fits into an individual lock. The bikes are locked until a payment is made at an automated kiosk [2.9]. The rental bike stands are designed to be the same height as the rental bikes, and cover the front wheels of the rental bikes.

Appendix C Source Extracts

[1] "UsabilityNet: Definition of usability", Usabilitynet.org, 2017. [Online]. Available: http://www.usabilitynet.org/management/b_what.htm. [Accessed: 04- Nov- 2017].

[2] Pictures of bikes, bike racks, and bike stands around UTSG. UTSG, Toronto, ON: University of Toronto, 2017.

Figure 2.1



Figure 2.2



Figure 2.3



Figure 2.4



Figure 2.5



Figure 2.6



Figure 2.7



Figure 2.8



Figure 2.9



Figure 2.10



[3] W. Toronto, "U of T Map", Map.utoronto.ca, 2017. [Online]. Available: <http://map.utoronto.ca/#>. [Accessed: 04- Nov- 2017].

Figure 3.1

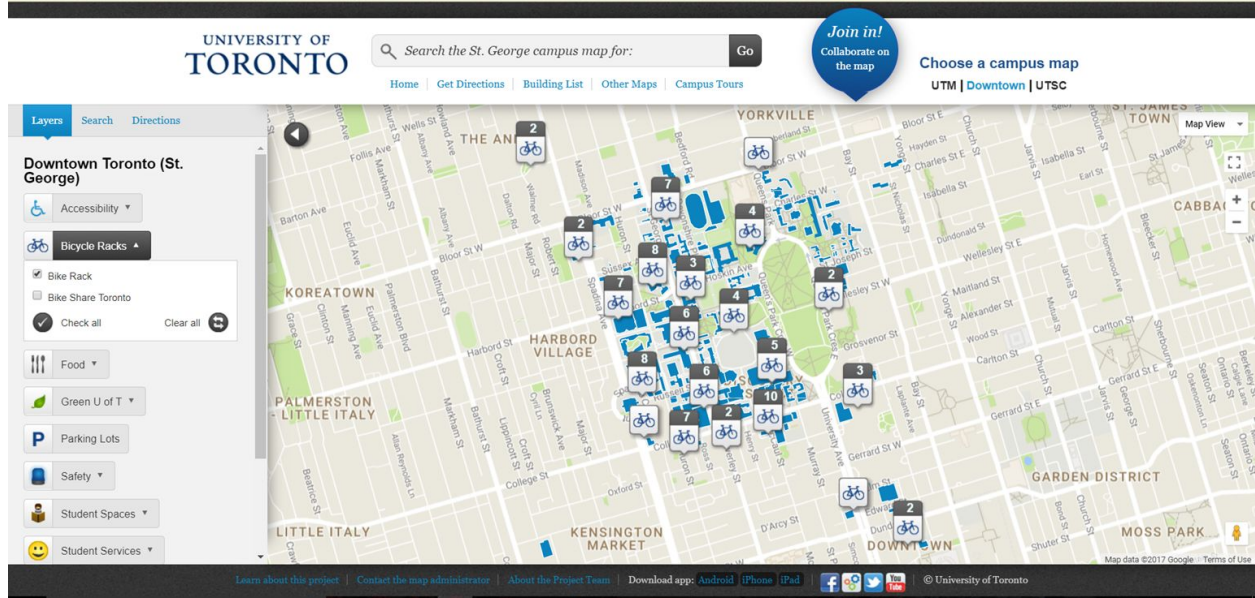
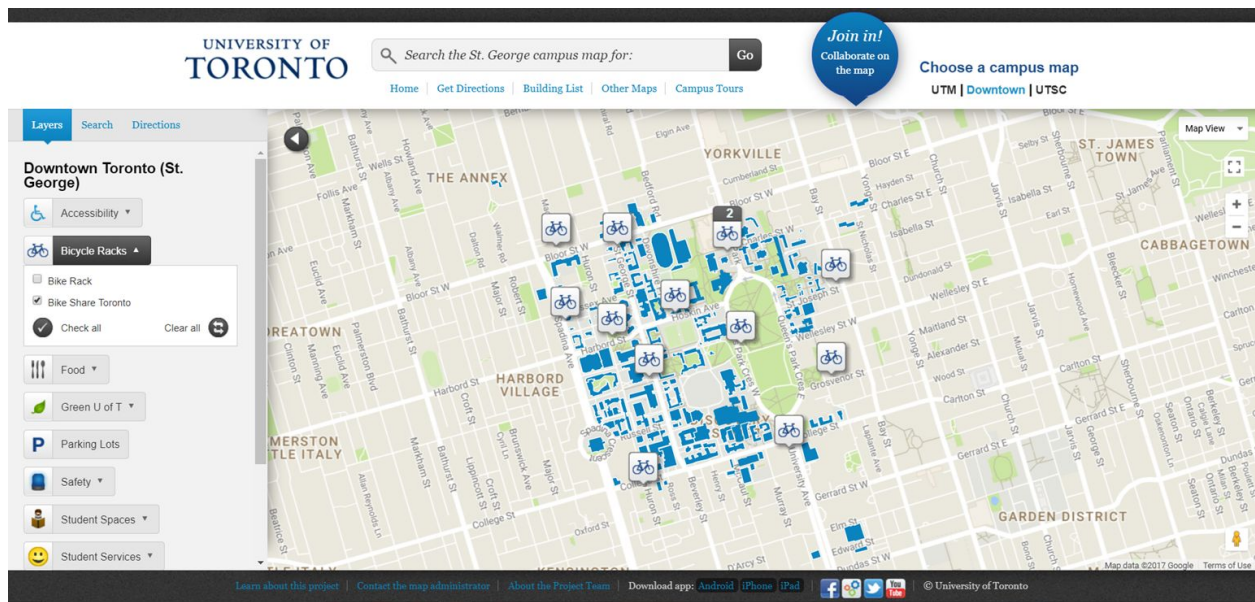


Figure 3.2



[4] "bikeparking.qxd", City of Toronto, 2017. [Online]. Available: https://www1.toronto.ca/city_of_toronto/transportation_services/cycling/files/pdf/bicycle_parking_guide.pdf. [Accessed: 04- Nov- 2017].

“Key items to consider when planning for bicycle parking

- 1) close proximity to an entrance
- 2) protecting bicycles from rain and snow
- 3) reducing chance of theft...

Short-term Parking—key points to consider

Short-term parking is usually provided at outside racks.

- **LOCATION**—Parking should be as close to a major entrance as possible. If it's too far away, cyclists will start locking their bikes to trees, posts, and street furniture. Use overhangs and awnings to provide shelter from weather. Be sure to choose a flat surface so that bicycles do not fall over, and avoid areas that will collect puddles.
- **VISIBILITY**—Parking should be easy to spot from the road. Cyclists will avoid parking in isolated areas—choose a visible, high-use area to deter thieves. Consider adding signage.
- **SPACE**—Make sure to leave enough room to lock bicycles side by side, even with panniers and wide handle bars. Rack manufacturers often overstate capacity. Plan for each bicycle to use up to .6 X 1.8 metres (2 feet by 6 feet). Allow at least .6m (2 foot) clearance between the rack and walls, poles, street furniture, motorists and pedestrians...

Choosing a Rack...

- When given the choice, cyclists will lock bikes to sign posts rather than to a poorly designed, poor quality rack. Good design is one that deters theft, by allowing the bicycle frame and one wheel (preferably two) to be locked to the rack.
- Make sure that cyclists can use both a cable and/or U Lock.
- Consider carefully before investing in vertical wall racks. They are a great way to manage in a tight space. Allow a horizontal dimension of at least .6 metres x 1.2 metres (2 feet x 4 feet) and a vertical dimension of 1.9 metres (6 foot, 4 inches). Many cyclists are discouraged, however, by having to lift their bicycles onto the wall so mount the rack as low as possible. If you are short on space, try to provide both horizontal and vertical racks
- Avoid racks that only hold the wheel. These racks can bend a bicycle's wheel. Also, only the wheel can be locked, making the bicycle an easy target for theft.
- Avoid racks that do not support the frame at two points—bicycles can fall over and the racks are often misused, resulting in reduced capacity.
- These are examples of recommended horizontal racks that allow at least one wheel and the frame to be locked to the racks, and offer two points of contact.

1. Post and Ring—You can request free post and rings on City-owned property. Use the online form at <http://www.city.toronto.on.ca/cycling/postandrings.htm> or call (416) 392-9253.

2. Inverted U.

3. Triangle.”

[5] "systems.pdf", Fietsparkeur.nl, 2017. [Online]. Available: <http://www.fietsparkeur.nl/normtekstengels.pdf>. [Accessed: 04- Nov- 2017].

[6] "The Best Bike Lock", The Best Bike Lock, 2017. [Online]. Available: <http://thebestbikelock.com/>. [Accessed: 04- Nov- 2017].

“Because of [U-locks’] rigidity they can be more challenging to transport than chain locks. And their limited size means you’ll find less things you can secure your bike to.”

Figure 6.1



Figure 6.2



Figure 6.3



Figure 6.4



Figure 6.5



Figure 7.1

Figure 7.2

THE BEST BIKE LOCKS										
All locks				U-locks		Chain locks		Tips & Tricks		How to lock your bike
Model	Thickness	Weight	Weight in cans of coke	Height	Width	Double bolted	In-house score	Sold Secure score	ART score	
Cocoweb Armbar	14 mm	3.2 lb (1.45 kg)	4	10.6" (26.8 cm)	6.7" (17 cm)	No	-	-	-	
OnGuard Bulldog Mini	13 mm	2.15 lb (0.98 kg)	2.5	3.55" (9 cm)	5.52" (14 cm)	Yes	63/100	-	-	
Kryptonite KryptoLok series 2 ATB	13 mm	3.10 lb (1.41 kg)	3.5	9.0" (22.9 cm)	5.0" (12.7 cm)	No	6/10	Silver	2/5	
OnGuard Bulldog ATB	13 mm	2.68 lb (1.215 kg)	3	9.06" (23 cm)	5.0" (12.7 cm)	Yes	63/100	-	-	
Abus Granit Strato 61 230	11 mm	2.39 lb (1.085 kg)	3	9.0" (23 cm)	5.0" (12.7 cm)	Yes	10/15	-	-	
Abus Granit Strato 61 300	11 mm	2.71 lb (1.229 kg)	3	12.0" (30 cm)	5.0" (12.7 cm)	Yes	10/15	-	-	
OnGuard Brute STD	16.8 mm	4.17 lb (1.86 kg)	5	7.96" (20.2 cm)	4.53" (11.5 cm)	Yes	97/100	Gold	-	
OnGuard Pitbull LS	14 mm	3.9 lb (1.77 kg)	4.5	11.5" (29.2 cm)	4.53" (11.5 cm)	Yes	80/100	Gold	-	
OnGuard Pitbull STD	14 mm	3.55 lb (1.61 kg)	4	9.0" (22.9 cm)	4.53" (11.5 cm)	Yes	80/100	Gold	-	
OnGuard Bulldog LS	13 mm	3.12 lb (1.42 kg)	3.5	11.5" (29.2 cm)	4.53" (11.5 cm)	Yes	63/100	Silver	-	
OnGuard Bulldog STD	13 mm	2.43 lb (1.10 kg)	3	9.06" (23 cm)	4.53" (11.5 cm)	Yes	63/100	Silver	-	

THE BEST BIKE 											
All locks U-locks Chain locks Tips & Tricks How to lock your bike About Blog											
Model	Thickness	Weight	Weight in cans of coke	Height	Width	Double bolted	In-house score	Sold Secure score	ART score		
Abus Granit Futura 64 150	11 mm	1.6 lb (0.726 kg)	2	6.0" (15 cm)	2.64" (6.7 cm)	Yes	11/15	-	-		
Kryptonite Evolution LITE Mini-6	11 mm	1.65 lb (0.74 kg)	2	6.0" (15.3 cm)	2.75" (7 cm)	No	7/10	Silver	-		
Kryptonite Messenger Mini	11 mm	1.83 lb (0.83 kg)	2	6.5" (16.5 cm)	3.75" (9.5 cm)	No	7/10	-	-		
Abus GRANIT Plus 640 150	12 mm	1.90 lb (0.86 kg)	2	6.0" (15 cm)	3.27" (8.3 cm)	Yes	12/15	Silver	-		
Kryptonite Evolution Mini-5	13 mm	1.94 lb (0.88 kg)	2.5	5.5" (14 cm)	3.25" (8.2 cm)	No	7/10	Silver	2/5		
Abus GRANIT Plus 640 230	12 mm	1.98 lb (0.90 kg)	2.5	9.0" (23 cm)	3.27" (8.3 cm)	Yes	12/15	Silver	-		
Abus Granit Futura 64 230	11 mm	2.05 lb (0.93 kg)	2.5	9.0" (23 cm)	2.64" (6.7 cm)	Yes	11/15	-	-		
OnGuard Bulldog Mini	13 mm	2.15 lb (0.98 kg)	2.5	3.55" (9 cm)	5.52" (14 cm)	Yes	63/100	-	-		
Abus U-mini 40	14 mm	2.16 lb (0.98 kg)	2.5	5.63" (14.3 cm)	3.15" (8.0 cm)	Yes	11/15	Silver	-		
Hiplok D	13 mm	2.20 lb (1 kg)	2.5	5.31" (13.5 cm)	2.76" (7 cm)	No	-	Silver	-		
OnGuard Bulldog Combo STD LM	11 mm	2.2 lb (1 kg)	2.5	9.06" (23 cm)	4.53" (11.5 cm)	Yes	58/100	-	-		

THE BEST BIKE 											
All locks U-locks Chain locks Tips & Tricks How to lock your bike About Blog											
Model	Thickness	Weight	Weight in cans of coke	Height	Width	Double bolted	In-house score	Sold Secure score	ART score		
Kryptonite New York Lock M18-WL	18 mm	6.1 lb (2.76 kg)	7	10.3" (26.1 cm)	3.9" (10.3 cm)	Yes	10/10	Gold & Motorcycle Gold	-		
OnGuard Brute LS	16.8 mm	5.07 lb (2.3 kg)	6	10.24" (26 cm)	4.37" (11.1 cm)	Yes	97/100	Gold	-		
Kryptonite New York Fahgettaboudit Mini	18 mm	4.55 lb (2.06 kg)	5.5	6.0" (15.3 cm)	3.25" (8.3 cm)	Yes	10/10	Gold & Motorcycle Gold	4/5		
Kryptonite New York Lock LS	16 mm	4.55 lb (2.06 kg)	5.5	10.25" (26 cm)	4.0" (10.2 cm)	Yes	9/10	-	-		
Kryptonite New York Lock Standard	16 mm	4.35 lb (1.97 kg)	5	8.0" (20.3 cm)	4.0" (10.2 cm)	Yes	9/10	Gold	4/5		
OnGuard Brute STD	16.8 mm	4.17 lb (1.86 kg)	5	7.96" (20.2 cm)	4.53" (11.5 cm)	Yes	97/100	Gold	-		
Kryptonite Evolution series 4 LS	14 mm	4.10 lb (1.86 kg)	5	11.5" (29.2 cm)	4.0" (10.2 cm)	Yes	8/10	Gold	3/5		
OnGuard Pitbull LS	14 mm	3.9 lb (1.77 kg)	4.5	11.5" (29.2 cm)	4.53" (11.5 cm)	Yes	80/100	Gold	-		
Kryptonite Evolution series 4 Standard	14 mm	3.75 lb (1.70 kg)	4.5	9.0" (22.9 cm)	4.0" (10.2 cm)	Yes	8/10	Gold	3/5		
Abus GRANIT X Plus 540 300	13 mm	3.65 lb (1.656 kg)	4	12.0" (30 cm)	4.25" (10.8 cm)	Yes	15/15	Gold	3/5		
Abus uGrip Plus 501	13 mm	3.64 lb (1.65 kg)	4	8.86" (22.5 cm)	3.74" (9.5 cm)	Yes	13/15	Gold	3/5		

[8] "Chain lock Comparison - The Best Bike Lock", The Best Bike Lock, 2017. [Online]. Available: <http://thebestbikelock.com/best-chain-lock/chain-lock-comparison/>. [Accessed: 04- Nov- 2017].

Figure 8.1

THE BEST BIKE LOCK									
All locks U-locks Chain locks Tips & Tricks How to lock your bike About Blog									
Model	Thickness	Weight	Weight in cans of coke	Length	Lock	In house score	Sold Secure score	ART score	
Kryptonite Evolution series 4 1055 Mini Integrated Chain	10 mm	4.40 lbs (2.00 kg)	5	21.5" (55cm)	Integrated	8/10	Gold	-	
Kryptonite KryptoLok series 2 955 Mini Integrated Chain	9 mm	3.90 lb (1.77 kg)	4.5	21.5" (55 cm)	Integrated	6/10	Silver	-	
Kryptonite Keeper 755 Mini Integrated Chain	7 mm	2.06 lb (0.93 kg)	2.5	21.5" (55 cm)	Integrated	5/10	Bronze	-	
Abus 1500 Web 60	4 mm	0.31 lb (0.14 kg)	0.5	24.0" (60 cm)	Integrated	3/15	-	-	
Abus 1500 60	4 mm	0.31 lb (0.14 kg)	0.5	24.0" (60 cm)	Integrated	3/15	-	-	
Abus Expedition-Chain 70/45/6 KS 65	6 mm	1.59 lb (0.72 kg)	2	26.0" (65 cm)	Padlock	6/15	-	-	
Hiplok Lite	6 mm	2.2 lb (1 kg)	2.5	29.5" (75 cm)	Integrated	-	Bronze	-	
Kryptonite New York Noose 1275	12 mm	6.95 lb (3.15 kg)	8	30.0" (75 cm)	Padlock	9/10	Gold	4/5	
Abus Combination Tresor 1385 75	7 mm	2.31 lb (1.05 kg)	2.5	30.0" (75 cm)	Integrated Combination	7/15	-	-	
Abus Catena 680 75	6 mm	1.87 lb (0.85 kg)	2	30.0" (75 cm)	Integrated	6/15	-	-	
Abus Catena 685 Shadow 75	6 mm	1.87 lb (0.85 kg)	2	30.0" (75 cm)	Integrated	6/15	-	-	

Figure 8.2

THE BEST BIKE LOCK									
All locks U-locks Chain locks Tips & Tricks How to lock your bike About Blog									
Model	Thickness	Weight	Weight in cans of coke	Length	Lock	In house score	Sold Secure score	ART score	
OnGuard Beast 8016L	14 mm	22.0 lb (9.98 kg)	25.5	72.0" (183 cm)	Padlock	99	-	-	
OnGuard Beast 8018	12 mm	13.7 lb (6.2 kg)	16	72.0" (183 cm)	Padlock	95	-	-	
OnGuard Mastiff 8019L	10 mm	10.5 lb (4.76 kg)	12	72.0" (183 cm)	Padlock	92	Gold	-	
OnGuard Mastiff 8021	10 mm	9.84 lb (4.46 kg)	11	72.0" (183 cm)	Integrated	78	Bronze	-	
Kryptonite New York Chain 1217	12 mm	13.15 lb (5.96 kg)	15.5	67.0" (170 cm)	Padlock	9/10	Gold	4/5	
Abus Ivy Chain 9100 170	10 mm	10.69 lb (4.85 kg)	12.5	67.0" (170 cm)	Integrated	13/15	Gold	3/5	
Abus Granit CityChain X-Plus 1060 170	10 mm	8.27 lb (3.75 kg)	9.5	67.0" (170 cm)	Integrated	15/15	Gold	3/5	
Abus CityChain 1010 170	9 mm	7.45 lb (3.38 kg)	8.5	67.0" (170 cm)	Integrated	12/15	Gold	-	
Kryptonite Evolution series 4 1016 Integrated Chain	10 mm	9.70 lb (4.40 kg)	11.5	63.0" (160 cm)	Integrated	8/10	Gold	-	
Kryptonite New York Legend Chain 1515	15 mm	15.95 lb (7.23 kg)	18.5	60.0" (150 cm)	Padlock	10/10	-	5/5	
Kryptonite New York Fahgettaboudit Chain 1415	14 mm	15.25 lb (6.92 kg)	17.5	60.0" (150 cm)	Padlock	10/10	Gold & Motorcycle Gold	-	

THE BEST BIKE LOCK										
All locks				U-locks		Chain locks		Tips & Tricks		How to lock your bike
Model		Thickness	Weight	Weight in cans of coke	Length	Lock	In house score	Sold Secure score	ART score	
OnGuard Beast 8016L		14 mm	22.0 lb (9.98 kg)	25.5	72.0" (183 cm)	Padlock	99	-	-	
Kryptonite New York Legend Chain 1515		15 mm	15.95 lb (7.23 kg)	18.5	60.0" (150 cm)	Padlock	10/10	-	5/5	
Kryptonite New York Fahgettaboudit Chain 1415		14 mm	15.25 lb (6.92 kg)	17.5	60.0" (150 cm)	Padlock	10/10	Gold & Motorcycle Gold	-	
OnGuard Beast 8018		12 mm	13.7 lb (6.2 kg)	16	72.0" (183 cm)	Padlock	95	-	-	
Kryptonite New York Chain 1217		12 mm	13.15 lb (5.96 kg)	15.5	67.0" (170 cm)	Padlock	9/10	Gold	4/5	
OnGuard Beast 8016		14 mm	12.13 lb (5.5 kg)	14	43.0" (110 cm)	Padlock	99	-	-	
Kryptonite New York Fahgettaboudit Chain 1410		14 mm	10.80 lb (4.90 kg)	12.5	39.0" (100 cm)	Padlock	10/10	Gold & Motorcycle Gold	-	
Abus Ivy Chain 9100 170		10 mm	10.69 lb (4.85 kg)	12.5	67.0" (170 cm)	Integrated	13/15	Gold	3/5	
Kryptonite New York Noose 1213		12 mm	10.55 lb (4.79 kg)	12.5	51.0" (130 cm)	Padlock	9/10	Gold	4/5	
OnGuard Mastiff 8019L		10 mm	10.5 lb (4.76 kg)	12	72.0" (183 cm)	Padlock	92	Gold	-	
OnGuard Mastiff 8021		10 mm	9.84 lb (4.46 kg)	11	72.0" (183 cm)	Integrated	78	Bronze	-	

[9] "Hal Breaks Locks", YouTube, 2017. [Online]. Available: <https://www.youtube.com/watch?v=kXXB8MqTNSU>. [Accessed: 04- Nov- 2017].

Appendix D:

Outdoor Bicycle Security at the University
of Toronto's Saint
George Campus

(See following pages)

Outdoor Bicycle Security at the University of Toronto's Saint George Campus

November 4, 2017

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Figure 1: Front of Robarts Library at 2017-10-30 11:00:00. Two bicycles are locked to the streetlight.

1 Introduction

Cycling is a popular mode of transportation for the frequenters of the University of Toronto's Saint George Campus (UTSG). As of November 2017 at UTSG, there are 91 official bicycle parking locations [1]. These numerous bicycle racks accommodate for cyclers among students and campus service providers.

Unfortunately, this popularity breeds theft of bicycles. UTSG campus police reports 66 bicycle-related thefts in 2016 [2]. UTSG's high number of bicycles permits bike-related robberies to occur.

This report investigates bicycle parking options at UTSG with respect to physical security, which is defined as the deterrence, protection, and recovery against stolen bikes in the form of physical safeguards and procedures [3] in the context of this report. It will consolidate field notes of the bicycle racks nearby Robarts Library and the involvement of campus police and other relevant stakeholders to holistically gauge bicycle security at UTSG.

2 Stakeholders

2.1 Campus Police

UTSG campus police enforce personal asset security laws on campus [4]. They are responsible for receiving and processing stolen bicycle reports from campus cyclists, recovering stolen bikes, and programs to reduce robbery rates.

2.2 UTSG Cyclists

A number of professors, teaching assistants, other campus service providers, and students choose to cycle to and from campus buildings. In general, they are interested in the protection of their bicycles against theft,

3 Bicycle Theft Occurences at UTSG

3.1 Bike Theft Statistics

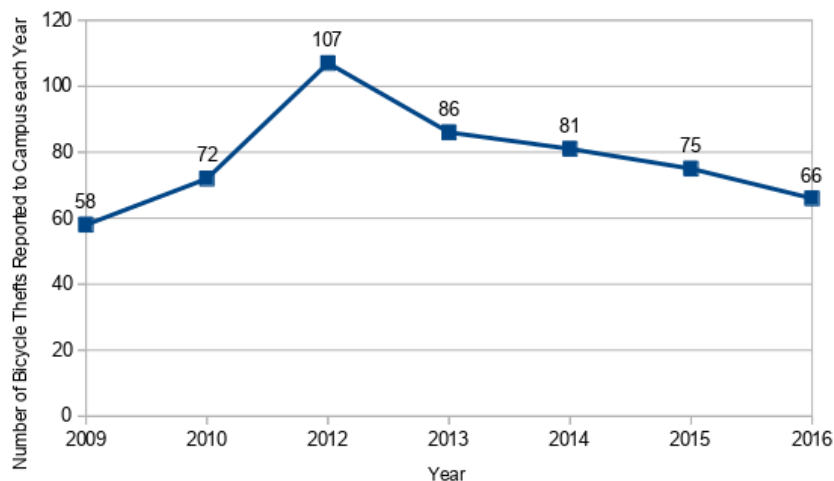


Figure 2: Yearly Bicycle Thefts at UTSG as reported by campus police [5, 6, 7, 8, 2].

As shown by Figure 2, the frequency of bike thefts have been steadily declining since a large spike in 2012. According to the 2013 UTSG campus police annual report, UTSG campus police repounded to the increasing number of bike thefts in 2012 with the Bait Bike program. Bait bikes, or decoy bicycles with GPS trackers meant to be stolen and traced back to bicycle thieves, led to the arrest of four bicycle thieves in late 2012 [6].

These is likely to have deterred thieves for fear of being caught causing the gradual decrease in bike thefts.

3.2 Evidence of Past Theft

3.2.1 Parked Bicycles Missing Parts

Figure 3 is evidence that individual bicycle parts are also stolen from secured bicycles in addition to entire bicycles. The frame is secured with a lock, but the front wheel was unscrewed and stolen. There are a number of bicycles with missing and damaged parts throughout campus, many of them being rusted like the bicycle in Figure 3. This suggests that bicycles missing parts are sometimes abandoned,



Figure 3: A U-locked bicycle with a missing wheel at UTSG.

3.2.2 Bike Theft Victims

First year UTSG student Rui Gao had his bicycle stolen on campus last month despite having a chain lock. He reported the incident to campus police, and has yet to locate his stolen bicycle [9].

4 Bicycle Parking System



Figure 4: Toronto City Cycling Committee's Standard Bike Post.

4.1 Bicycle Parking Locations at UTSG

The bicycle racks in front of and near Robarts Library are the focus of this inspection due to the high number of bicycle racks located there [1]. Despite being easily seen, public exposure is insufficient to deterring theft. A Toronto Star reporter successfully stole their own bicycle with bright orange bolt cutters in front of Robarts Library at noon [10]. This indicates that public robberies are difficult to distinguish from malfunctioning locks, so bystanders cannot deter thieves.

4.2 Personal Locks

The police recommend using two different locks at the same time to make robbery more difficult [11], but most bicycles seen had only one.

4.2.1 U-Lock

These locks consist of a rigid U-shaped metal bar secured at both ends. The locks are hooked around a single bicycle and a suitable parking location discussed in section 4.3. They are difficult to break, but do not sufficiently prevent thievery. Figure 3 shows a U-locked bike with a wheel removed. They are often too small to secure all parts of the bike.

4.2.2 Cable Lock

These locks are often long steel cables wrapped around all bicycle parts and a post. More parts of a bicycle can be secured with a cable lock than a U-lock, but they are proven breakable by Rui's chained bicycle being stolen [9].

4.3 Bicycle Parking Locations

4.3.1 Toronto Cycling Committee's Ring Post

The most predominant bicycle rack at UTSG is the Toronto City Cycling Committee's ring-shaped post shown in Figure 4. The post is a 7 cm diameter metal rod through the centre of a 40 cm ring, 25 cm off the ground. Its base is embedded in concrete. A lock is placed around the bicycle and the post to secure the bicycle. It complies with most of the FietsParKeur's Standards for Bicycle Parking Systems, including the locks being fastened at least 25 cm off the ground, and is likely to not yield under the abuse given by a 25 kg dynamic mass due to being made of metal and embedded in concrete [12]. Personal attempts to shift or bend the post with significant force were unsuccessful.

4.3.2 Wheel Racks

They are two 2 m parallel metal bars welded to 25 cm high hoops acting as racks for bicycle wheels. A personal lock affixes the bicycle to the rack. These are less secure than the ring posts discussed in Section 4.3.1 since they are not secured to any surface. Figure 5 shows the rack with bikes



Figure 5: Wheel rack securing bike wheels being lifted slightly.

locked to it being lifted slightly off the ground by hand. The City of Toronto’s guide to Bicycle Parking asserts that thieves have stolen entire racks in the past [11], meaning secured bicycles can be taken with the rack. In addition, locks securing bicycle wheels to the stand can lie less than 25 cm from the ground, which is incompliant with anti-theft construction outlined in FietsParKeur bicycle parking system standards [12] since locks can be smashed against the ground and broken. Shorter locks can only secure the wheel in the rack, meaning the rest of the bike can be stolen by removing the secured wheel.

5 Conclusion

Security systems against bicycle theft at UTSG are not sufficient for UTSG cyclists and campus police. Public bicycle parking provide little social protection against theft, locks and posts used on campus are easily compromised, and stolen bicycles leave nearly no evidence helping campus police recover stolen bicycles.

References

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- [3] L. Fennelly, *Physical Security: 150 Things You Should Know*. Elsevier, November 2016.
- [4] University of Toronto. Campus Police. Date Accessed: 2017-11-04. [Online]. Available: <http://campuspolice.utoronto.ca/>
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- [7] —, “Annual report,” 2014. [Online]. Available: <http://campuspolice.utoronto.ca/wp-content/uploads/2017/04/2014-Campus-Police-Annual-Report-University-Affairs-Board.pdf>
- [8] —, “Annual report,” 2015. [Online]. Available: <http://campuspolice.utoronto.ca/wp-content/uploads/2017/04/2015-Campus-Police-Annual-Report-University-Affairs-Board.pdf>
- [9] “Interview with Rui Gao,” 2017.
- [10] J. Tapper, “Bike theft revealed: Star hits the streets to test how easy it is to steal a locked-up bike,” *Toronto Star*, 2012. [Online]. Available: https://www.thestar.com/news/gta/2012/03/20/bike_theft_revealed_star_hits_the_streets_to_test_how_easy_it_is_to_steal_a_lockedup_bike.html
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- [12] FietsParKeur, “Standards for bicycle parking systems,” 2003. [Online]. Available: https://www1.toronto.ca/city_of_toronto/transportation_services/cycling/files/pdf/bicycle_parking_guide.pdf

Appendix D Source Extracts



[1]

1. WHAT IS PHYSICAL SECURITY?

There are different opinions and interpretations about exactly what physical security is, but the two definitions below incorporate all the aspects and are all-inclusive.

The American Society for Industrial Security (ASIS) International (2005) defines *physical security* as follows:

Physical security focuses on the protection of people, property, and facilities through the use of security forces, security systems, and security procedures. Physical security personnel oversee proprietary or contract uniformed security operations, identify security system requirements, assess internal and external threats to assets, and develop policies, plans, procedures, and physical safeguards to counter those threats. Physical security can include the use of barriers, alarms, locks, access control systems, protective lighting, CCTV, and other state-of-the-art security technology.¹

The Army Field Manual, No. 3–19.30 defines *physical security* as “that part of security concerned with physical measures designed to safeguard personnel; to prevent unauthorized access to equipment, installations, material, and documents; and to safeguard against espionage, sabotage, damage, and theft.”²

[3]

- **SPACE**—Make sure to leave enough room to lock bicycles side by side, even with panniers and wide handle bars. Rack manufacturers often overstate capacity. Plan for each bicycle to use up to .6 X 1.8 metres (2 feet by 6 feet). Allow at least .6m (2 foot) clearance between the rack and walls, poles, street furniture, motorists and pedestrians.
- **SECURITY**—Thieves have been known to carry off entire racks full of bikes. Racks should be embedded in cement or affixed with tamper-proof bolts or screws. Make sure the area is well-lit at night. Consider adding a security camera and signage to advertise it. Make sure building security patrol the racks throughout the day.



X This rack is not being used because it is up against the wall and is partially obstructed by a concrete barricade.

The Lock

- Invest in the best quality lock (or two) that you can afford—usually a hardened steel U lock or hardened steel chain and padlock.
- Use two different locks—one for each wheel—so that a thief would need two different types of tools.
- Make sure chains cannot be smashed against the ground.
- Position a U lock so that the keyhole faces down towards the ground. A keyhole that is located in the middle of the straight bar (instead of at the end) offers greater security.

Appendix B: Source Extracts

[1] Transcript of Interview with Campus Police
November 27, 2017

“Drop bike, they have a bike that’s kinda a garbage bike, they ride it up to another bike to make it look as their locking their own bike but they’re really cutting the lock of the other bike so they leave their drop bike there and they leave with the new bike, so it’s really hard, cause if we have information about bike thefts going on, we [wear] plain clothes and watch, and sometimes we almost miss them.”

“These guys know what they’re doing, so you have your low level guys doing it, and you have more professional ones, guys that drive in a van, there’s a team of them, cause there’s a lot of money, a lot of these guys bring a lot of nice bikes, so you get your different level of guys stealing them. Students, probably, I think over the years there have been some”

“How much money can you get for a bike?”

“It depends, the guys that are stealing it for drugs will take whatever they can get, but some people they sell them on Kijiji. Depending on the bike that you stole you can earn 100, 200, 300 bucks.... People will see what they think is their bike on Kijiji and they’ll call us , well that’s why it’s important to register your bike with the police, know your serial number. We had someone have their bike stolen and it’s locked up somewhere else. Well you have to prove its your bike, ‘well it’s my bike I can tell by looking at it’, well we can’t just cut the lock and give it to you we need proof. So the bike being registered is a way of proving it, knowing your serial number. You’ll be surprised, bike theft is a pretty big business. Depends on how you lock it too, if you lock it with a cable lock, your bike is gonna be gone. If you lock it with a ULOCK, it’s gonna be harder to do”

“Do you find that most bike thefts are due to poor locking?”

“Yep”

“So a majority of all of them?”

“A majority. The cable is the easiest. If he has cable cutters, any cable lock is gonna be cut. The ULocks are harder to do. We’ve had guys over the years that are prepared with small car jacks, they put them in the u lock and they can snap them. People ask ‘Do bikes go missing at more at nighttime than daytime, not rly. A lot of times people do it during day time when it’s busy. We’ve been sitting day time in plain clothes in cars and have almost missed guys doing that. You have to pay attention to the bikes the bring and what they leave on.

“If people knew how to lock their bikes properly, you would save a lot more time dealing with this”

“I think it can bring it down but in reality, anything that can be locked can be unlocked, so they’ll just come up with another way to do it. If you lock your ULock downwards, instead of having the keylock facing up, bc they’re putting acid or frozen in the lock and just banging it.”

“What about bike where just one wheel is missing?”

“That’s different, that’s just guys doing parts and there’s some parts on some parts that are worth money. We even have people caught that steal bikes bc they wanted a ride home. That’s slowed down a little bit now bc with these new bikes you can rent.”

“The students down here have cheaper bikes, worth anywhere from 100 to 200 dollars... A lot of people don’t even report missing bikes cause they just go buy a cheap one. I know some universities have actual bike cages so you can put your bike in an actual bike cage and you lock it... You definitely have to consider cost. These students get a 20 dollar lock from Canadian Tire, and that’s why there locks are so easily broken.”

“In the summer, we have our officers on bikes so it’s easier to catch these thefts, but yeah it definitely takes more time and effort to deal with this when we could be using these resources to deal with other stuff. Stolen laptops and other accessories is a big one”

[2] UTSG Campus Police Services, “Annual report,” 2016. [Online]. Available: <http://campuspolice.utoronto.ca/wp-content/uploads/2017/09/2016-Campus-PoliceAnnual-Report-University-of-Toronto-Affairs-Board.pdf>

REPORTABLE INCIDENTS

2016 Statistical Overview - St. George Campus

PROPERTY CRIMES	2016	2015	16vs15
Arson	1	1	0
Attempt Theft	3	3	0
Break and Enter	9	12	-3
Unlawfully in a Dwelling	1	0	1
Fraud/ False Pretences	6	3	3
Mischief over \$5000 Univ.	0	2	-2
Mischief under \$5000	112	140	-28
Mischief - Interfere lawful enjoyment of property	1	0	1
Theft over \$5000.00	4	5	-1
Theft under \$5000.00	228	297	-69
Theft under \$5000.00 Bicycles	66	75	-9
Possession Stolen Property	0	0	0
Classification Total	431	538	-107

[3] L. Rundle, "To Catch a (Bike) Thief", *U of T Magazine*, 2006.

“Bike Bait works just like you might guess: an undisclosed number of bicycles (the bait) are planted around campus with a “very, very well-hidden” GPS (or global positioning system) beacon, says program co-ordinator Cpl. Peter Franchi. This beacon allows police to track the bike – and reel in the crook. U of T is the second Canadian university to implement the program, which is modelled after a Victoria Police Department initiative that reduced thefts by almost 20 per cent in six months. U of T’s

other campuses and the City of Toronto are both watching the program closely. “Depending on our success,” says Cpl. Franchi, “others may adopt the program.”

[4] Krug, S. (2006). *Don't make me think!*. 2nd ed. Berkeley, CA: New Riders, pp.6, 36.

“If it’s short, it’s more likely to actually be used. There’s a good usability principle right there: if something requires a large investment of time—or looks like it will—it’s less likely to be used.” (pg 6)

“If you’re not going to use an existing Web convention, you need to be sure that what you’re replacing it with either (a) is so clear and self-explanatory that there’s no learning curve—so it’s as good as a convention, or (b) adds so much value that it’s worth a small learning curve. If you’re going to innovate, you have to understand the value of what you’re replacing, and many designers tend to underestimate just how much value conventions provide. My recommendation: Innovate when you know you have a better idea...” (pg 36)

[5] City of Toronto, "Guidelines for the Design and Management of Bicycle Parking Facilities", Toronto, 2017.

“Short-term or “visitor” bicycle parking is designed to be used for a few minutes up to a couple of hours. As a result, short-term parking should be easily accessible, racks should provide a secure point for locking up and it is best if racks are covered for weather protection.”

“Two points of contact between the bicycle and the rack. This allows both the frame and at least one wheel to be locked to the rack and supports the bicycle.”

“Design Options: Materials to look for: galvanized steel; industrial grade materials. Materials to avoid: wood; soft metals; untreated metals that will rust; and cast composites that are brittle and may crack under impact. For rack designs that have welded sections avoid materials like stainless steel that do not weld strongly as weak welded sections can easily be broken by thieves or vandals. Finishing: Racks should have a smooth outer surface that will not damage or scratch bicycle frames.”

2.7.3 Uses with High Bicycle Parking Demand

Some developments have the potential to generate a higher than average demand for bicycle parking.

Examples include:

- Schools – elementary, high schools, colleges and universities – Many students do not drive or do not own a personal motor vehicle.
- Hospitals, large factories and major transit hubs are destinations for large numbers of visitors, customers and employees who may require access to the site at all hours for shift work, late night travel schedules or emergencies.
- Places of assembly – stadiums, community centres or concert halls – can receive a very large number of short-term visitors during special events. These locations can create a high demand for secure bicycle parking.
- Places of worship like places of assembly, receive a considerable number of visitors at one time thereby increasing the demand for secure bicycle parking.



Bicycle Parking at Robarts Library, University of Toronto. The University's St George Campus has bicycle parking near the entranceways of most of its major buildings.

[6] Manifesto, *the Bike Hanger 001*. 2011.

[7] Manifesto, *Picture of Manifesto Bike Hanger in Seoul, Korea*. 2011.

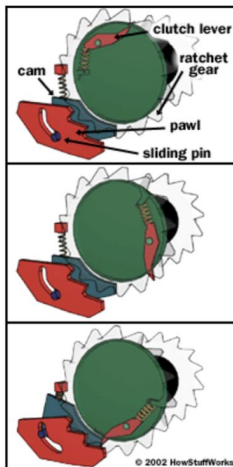


[8] “TrackR Bravo is the world’s thinnest tracking device,” *Interesting Engineering*, 02-Dec-2017. [Online]. Available:

<https://interestingengineering.com/trackr-bravo-is-the-worlds-thinnest-tracking-device>. [Accessed: 03-Dec-2017].



[9]T. Harris, “How Seatbelts Work,” *HowStuffWorks*, 09-Apr-2002. [Online]. Available: <https://auto.howstuffworks.com/car-driving-safety/safety-regulatory-devices/seatbelt4.htm>. [Accessed: 02-Dec-2017].



The second kind of system locks the spool when something **jerks the belt webbing**. The activating force in most designs is the speed of the spool rotation. The diagram shows a common configuration.

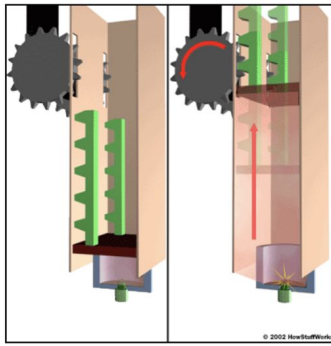
The central operating element in this design is a **centrifugal clutch** -- a weighted pivoting lever mounted to the rotating spool. When the spool spins slowly, the lever doesn't pivot at all. A spring keeps it in position. But when something yanks the webbing, spinning the spool more quickly, **centrifugal force** drives the weighted end of the lever outward.

The extended lever pushes a **cam** piece mounted to the retractor housing. The cam is connected to a pivoting pawl by a sliding pin. As the cam shifts to the left, the pin moves along a groove in the pawl. This pulls the pawl into the spinning ratchet gear attached to the spool. The pawl locks into the gear's teeth, preventing counter-clockwise rotation.

In some newer seatbelt systems, a **pretensioner** also works to tighten the belt webbing. In the next section, we'll see how these devices work.

The Pretensioner

« PREV NEXT »



When the gas is ignited, the pressure pushes the piston up to rotate the retractor.

The idea of a pretensioner is to tighten up any slack in the belt webbing in the event of a crash. Whereas the conventional locking mechanism in a retractor keeps the belt from extending any farther, the pretensioner actually **pulls in on the belt**. This force helps move the passenger into the optimum crash position in his or her seat. Pretensioners normally work together with conventional locking mechanisms, not in place of them.

There are a number of different pretensioner systems on the market. Some pretensioners pull the entire retractor mechanism backward and some rotate the spool itself. Generally, pretensioners are wired to the same central control processor that activates the car's [air bags](#). The processor monitors mechanical or electronic **motion sensors** that respond to the sudden deceleration of an impact. When an impact is detected, the processor activates the pretensioner and

then the air bag.

[10]"Who's trying to steal your bike (and what tools are they using)? - The Best Bike Lock", *The Best Bike Lock*, 2017. [Online]. Available: <http://thebestbikelock.com/best-bike-lock/who-is-trying-to-steal-your-bike/>. [Accessed: 03- Dec- 2017].

3. Hacksaws



Just like a pair of cable cutters, a hacksaw is cheap and pretty easy to carry around unnoticed. And just like the cable cutters it will slice through a cable lock very easily. With enough time, a hacksaw will also cut through low quality [U-locks](#) and [chains](#) which tend to be made from thin, soft metal.



The biggest bolt cutters (36 and 42") are usually only used by serious bike thieves. But they can crop mid range U-locks and chains and sometimes even the higher end locks.

To guarantee that a lock cannot be cropped by *any* manual bolt croppers you need a lock made from good quality, properly hardened steel with a diameter of 16 mm.

But bear in mind not everyone with 36 or 42" bolt cutters can crop a high security locks that's less than 16 mm thick. It takes a thief with considerable skill (and weight!) who is able to maneuver the lock into a favorable position where they can use the ground for leverage. Again, [good locking technique](#) can minimize this risk.

[11] Notes of interview with bike shop, Bikes on Wheels staff . December 3, 2017

Student: "We are designing for a new bike lock system to prevent bike theft. This is our prototype, a retractable chain station. How this system work is that you basically take the retractable chain from the stand and then wrap it around your bike. We also set an alarming system that get triggered with the chain is cut or the lock is broken. Do you have any concerns of certain part?"

Staff: "Well, I wonder where the chain can be attached to. Is it just the front wheel?"

Student: "No, it can wrap around the whole bike and then the end is attached to a small lock on the other end."

Staff: "Okay, I see. It is pretty good. It just depends on how easy the chain can be cut. Everything else looks secured. "

Student: "We are using a chain that is about 16 mm in diameter, do you think it is enough?"

Staff: "16 mm is good."

Student: "One last question: what do you think is the most targeted part of the bike."

Staff: " I'd say the wheels. That's why I asked you if the system only locks the front wheel. And handles, if accessible. If the lock, cable, chain is easy to cut, then they'll take the whole bike"

[12]"PRESTO Devices and How to Use Them," *PRESTO*. [Online]. Available: <http://getpres.to/en/presto-devices-and-how-to-use-them/>. [Accessed: 02-Dec-2017].

Fare Payment Device

Located on all surface vehicle (streetcars and buses) entrances and on a PRESTO-enabled subway station turnstile in select subway station entrances.

As you enter, tap your card against the device to pay for your ride. Your PRESTO card acts as your Proof-of-Payment (POP) after tapping to board.



[13] "Velodome Shelters - Wire Bike Lockers", *Velodomeshelters.com*, 2017. [Online]. Available: <http://www.velodomeshelters.com/wire-bike-lockers.html>. [Accessed: 02- Dec- 2017].

Wire Bike Lockers

Velodome's Wire Bike Lockers are a space efficient, vertical bicycle storage solution for facilities that want to provide an extra level of security for bicycles.

Each locker is only 40" wide x 52" deep x 98"high and comes standard with one MaxRack bike hanger. An additional MaxRack is an option to store 2 adult bicycles... with plenty of storage space to spare. This flexible locker system expands with modular add-on sections and can be configured with and without a top, side or rear grids.

- Modular
- 6 gauge or 10 gauge, 2" square, welded wire mesh
- Security- Mortise cylinder, keypad or padlock
- Customization available upon request

Ideal for:
Bike Rooms • Parking Garages • Basements
Apartment Buildings • Commercial Buildings
Dormitories



Modular Lockers

ColorsViewsSpecsOptionsCapacity

Specifications

Frame: 1" x 1" steel tube

Wire Mesh: 2" x 2" grids. 6 or 10 gauge wire

Finish: Powder coated grey (standard)

Dimensions:

Model	Length	Width	Height
Bike Locker (single)	52"	40"	98"

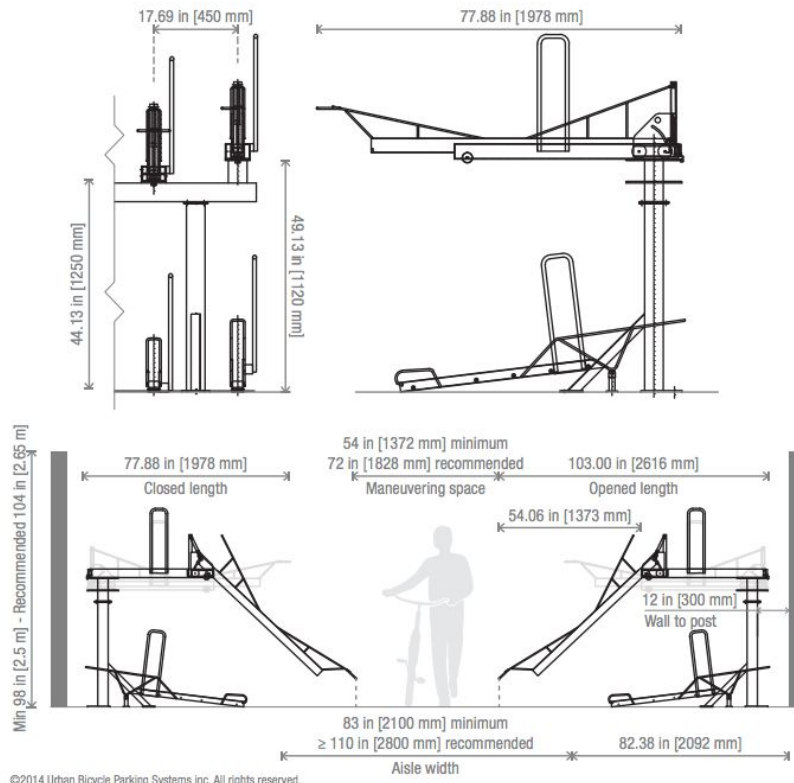
CADdetails

[14] Urban Racks, *Urban Double Stacker Rack*. 2014.



Urban Double Stacker

Model: UB-2500-SM-2WG



Specifications

Capacity	
Bicycles per set	2 (one up and one down)
Bicycle spacing	17.69 in [450 mm]
Weight	
Per two bicycle spaces	± 89.65 lbs [40.75 kg]
Materials	
Assembly material	Steel
Available finishes	
Powder coated (grey)	

These drawings are not for construction purposes and are for information purposes only. All information contained herein was current at the time of development but must be reviewed and confirmed by Urban Racks to be considered accurate.

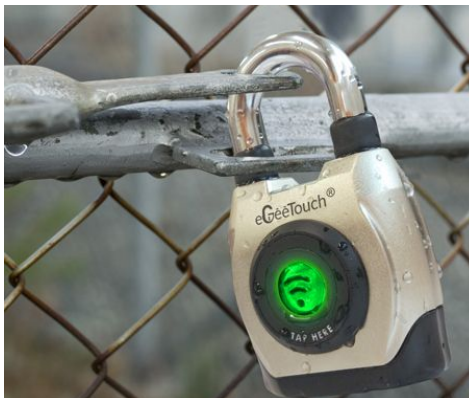
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1-888-717-8881 sales@urbanracks.com

For more product and company information, please visit us at www.urbanracks.com

[15] eGeeTouch, *Weatherproof, Outdoor Smart Padlocks*. 2017.



[16] W. Toronto, "Bicycle Racks | U of T Map", *Map.utoronto.ca*. [Online]. Available: <http://map.utoronto.ca/access/bicycle-racks>. [Accessed: 02- Dec- 2017].



Bicycle Parking: Banting Institute

3 post & ring bicycle stands.

Address :

Associated Building : [Banting Institute](#)

[Map it](#)

Numbers of every bike post on campus was given, and I added them up to get ~1600.

[17] How Silent Alarms Work : HomeSecurityResource.org", *Homesecurityresource.org*. [Online]. Available: <http://www.homesecurityresource.org/how-silent-alarms-work/>. [Accessed: 03- Dec- 2017].

“The Details of Silent Alarms

Using silent alarm monitoring, you’ll have the option to press a panic button to activate the alarm. You may also have a duress code that you can enter. When a silent alarm is tripped, your home security provider will be alerted to the emergency event in your home, and the authorities will be notified and sent to your home.

When using a panic button or duress code, home security systems do not require a follow up phone call or other confirmation to determine whether or not there is a true emergency: you’ve already told them that’s the case. Instead of calling your phone first, your home security monitoring station will immediately pass the alert on to the authorities, saving you valuable moments.”

[18] Lisa Li, private images, Oct. 2017.

